

Sleep, Rhythms, Learning, and Memory

Sleep and Dreams: April 29, 2026. H. C. Heller



- Why should you care?
- Types of memory
- Human experiments on influences of
 - Sleep and circadian rhythms on memory
- Animal experiments to discover mechanisms.
- Animals from the north, animal models of learning disability

Why should you care about sleep rhythms and memory?



> *J Biol Rhythms*. 2024 Feb;39(1):5-19. doi: 10.1177/07487304231207330. Epub 2023 Nov 18.

The Negative Effects of Travel on Student Athletes Through Sleep and Circadian Disruption

H Craig Heller¹, Erik Herzog², Allison Brager³, Gina Poe⁴, Ravi Allada⁵, Frank Scheer⁶, Mary Carskadon⁷, Horacio O de la Iglesia⁸, Rockelle Jang⁴, Ashley Montero⁹, Kenneth Wright¹⁰, Philippe Mouraine¹¹, Matthew P Walker¹², Namni Goel¹³, John Hogenesch¹⁴, Russell N Van Gelder¹⁵, Lance Kriegsfeld¹², Cheri Mah¹¹, Christopher Colwell¹⁶, Jamie Zeitzer¹¹, Michael Grandner¹⁷, Chandra L Jackson^{18, 19}, J Roxanne Prichard²⁰, Steve A Kay²¹, Ketema Paul²²

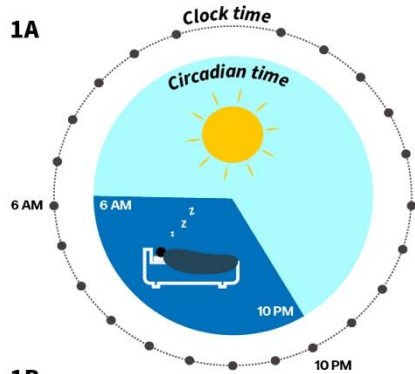
Affiliations + expand

PMID: 37978840 DOI: 10.1177/07487304231207330

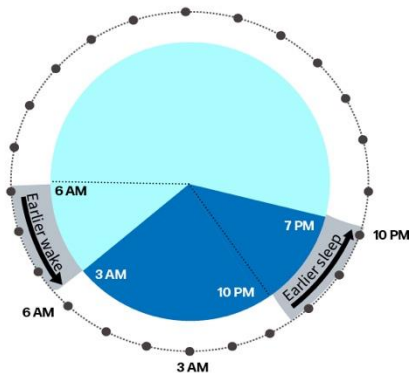


Up Close and Personal with Jet Lag

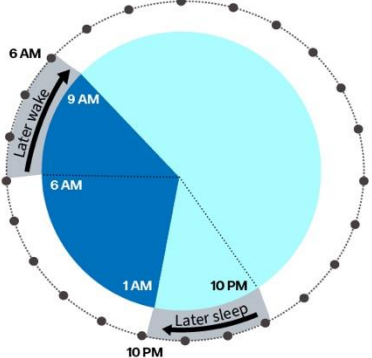
1A



1B



1C



Normal bedtime 10 PM, normal wake time 6 AM

Traveling east 3 time zones:

Bedtime 7 PM home time -- Bad sleep

Wake up 3 AM home time – not alert

Morning competition 6 AM home time?

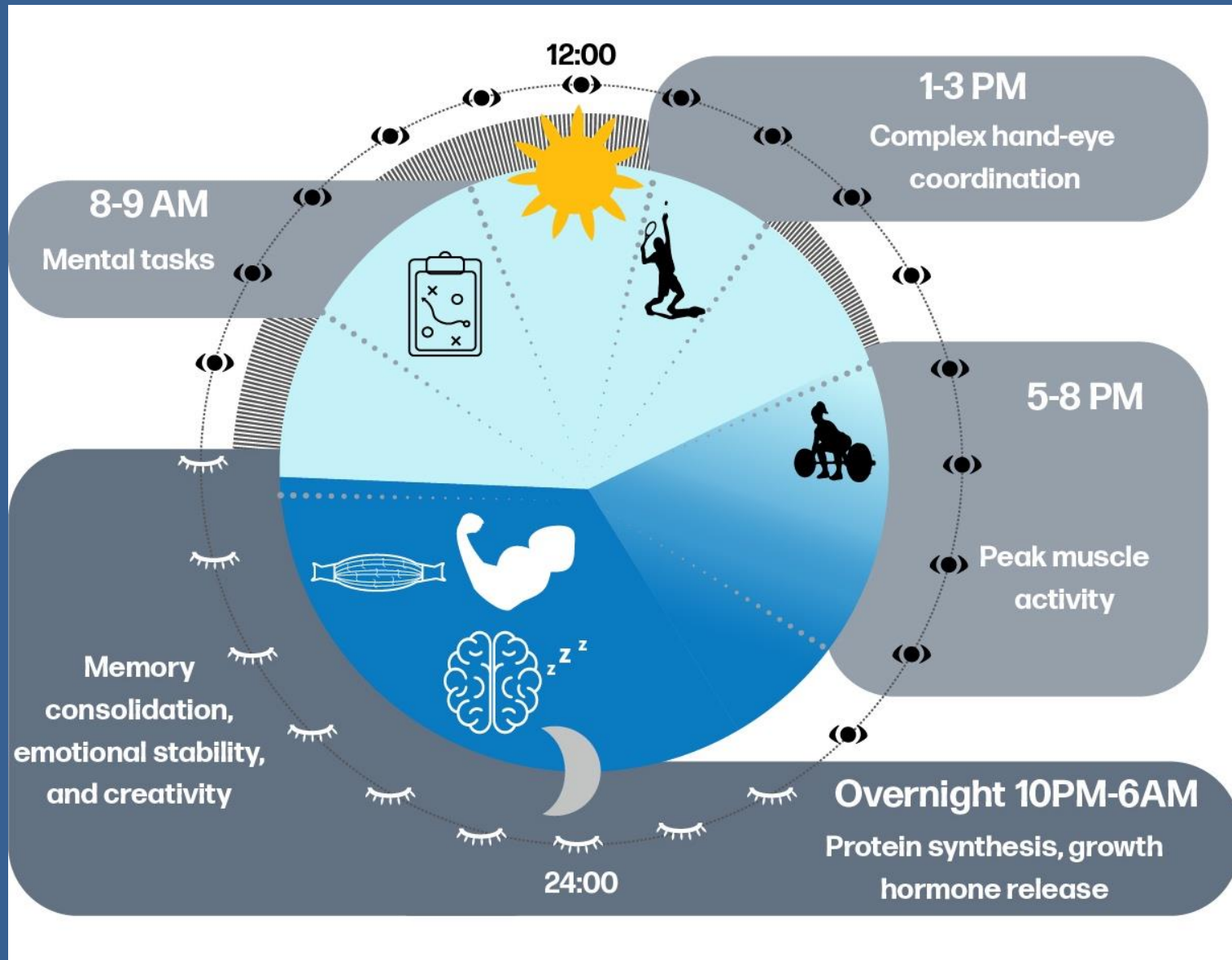
entrainment

Traveling west 3 time zones

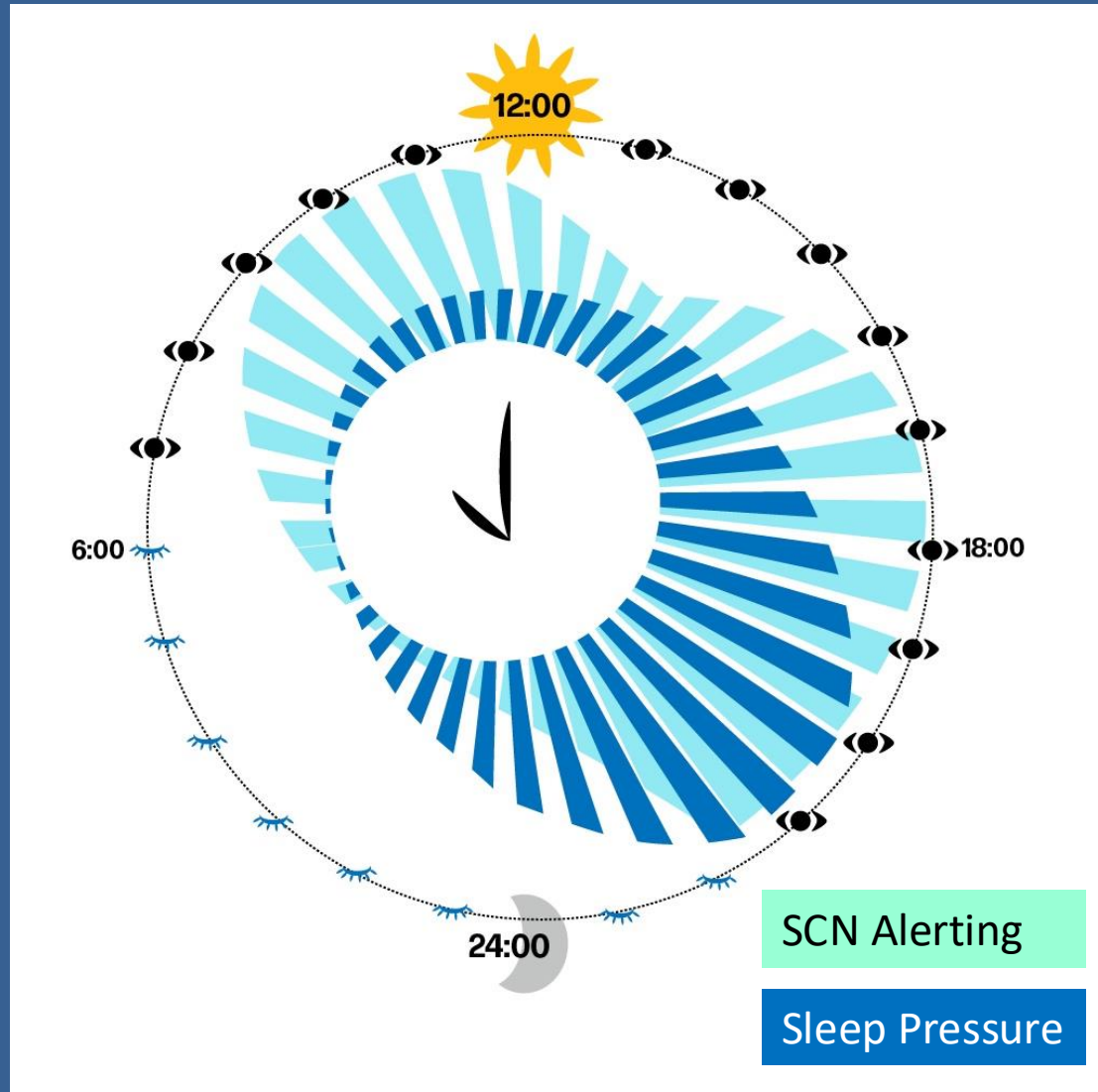
Home Bedtime now 1AM eastern time

Home Wake time is now 9 AM eastern time,
but 6 AM home time. – morning classes/exams
at 7 AM circadian time.

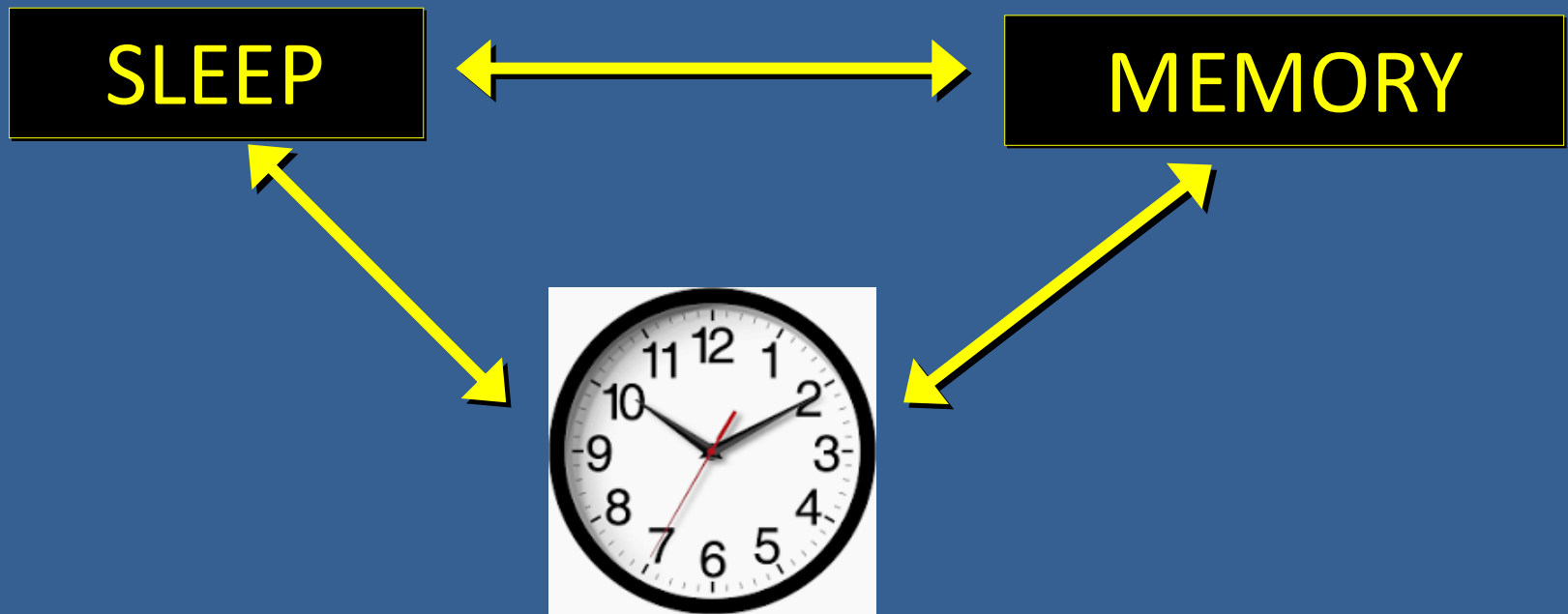
Effects of Time and Circadian Misalignment on Performance



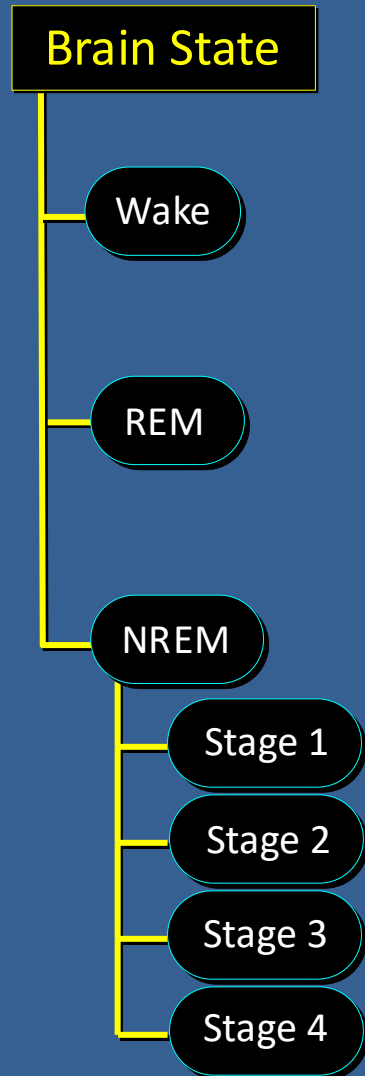
Interaction of the clock and the homeostat



The Question For Today



The question is more complex than it may seem.



Brain State

Wake

REM

NREM

Stage 1

Stage 2

Stage 3

Stage 4

Memory Types

Declarative

Episodic

Semantic

Non-Declarative

Procedural

Implicit

Non -
associative

Conditioning

Memory Stages

Acquisition/
Encoding

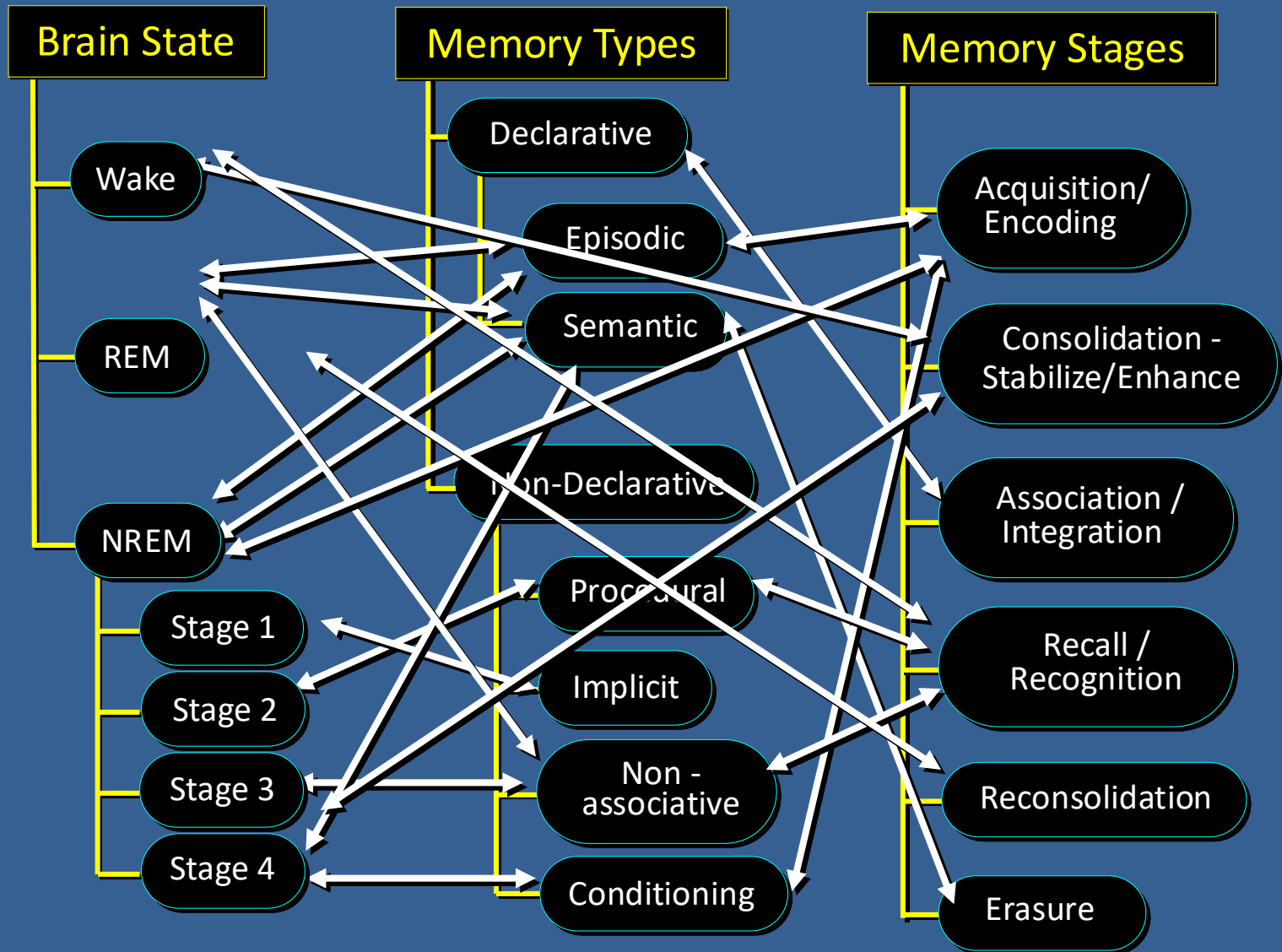
Consolidation -
Stabilize/Enhance

Association /
Integration

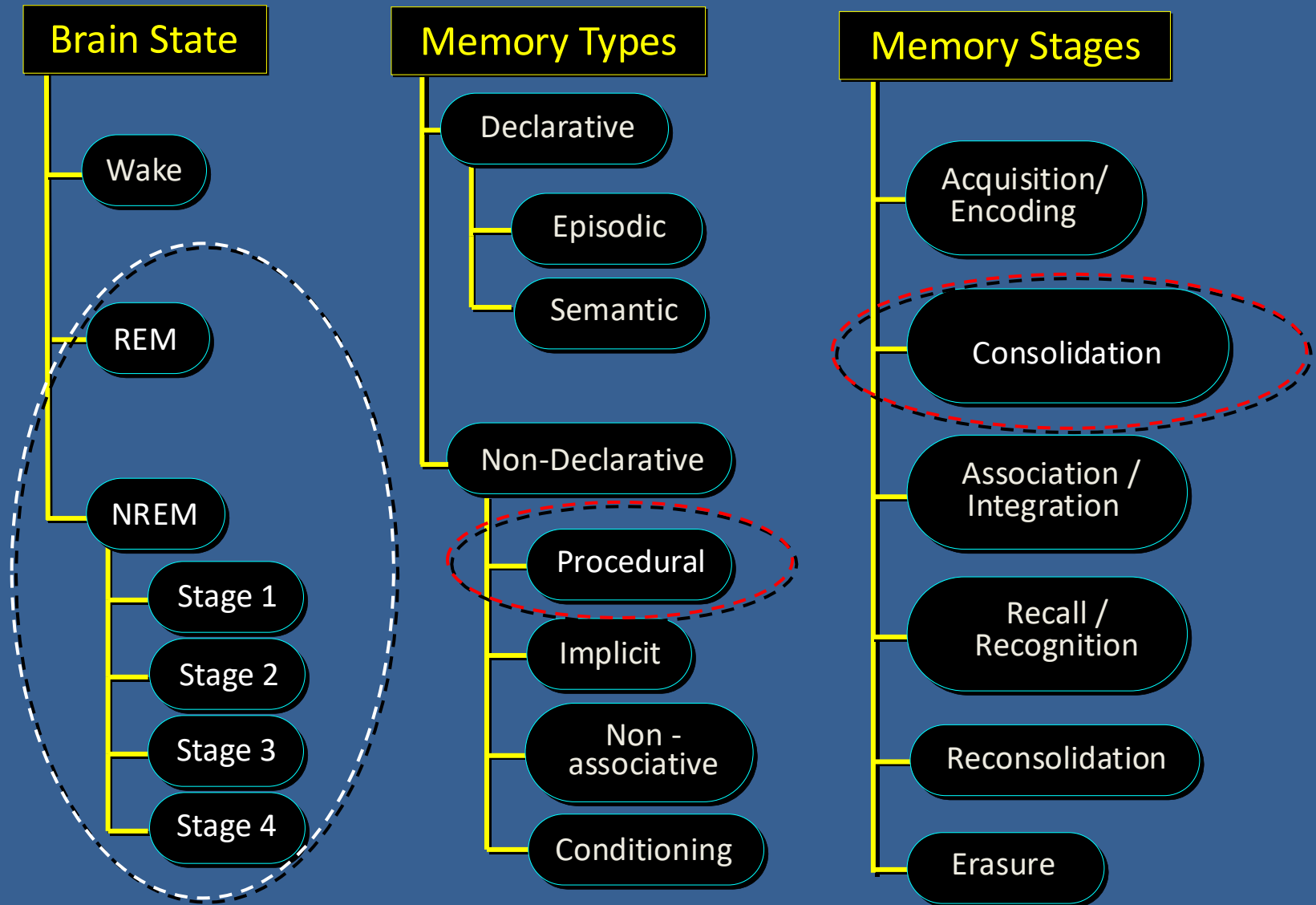
Recall /
Recognition

Reconsolidation

Erasure

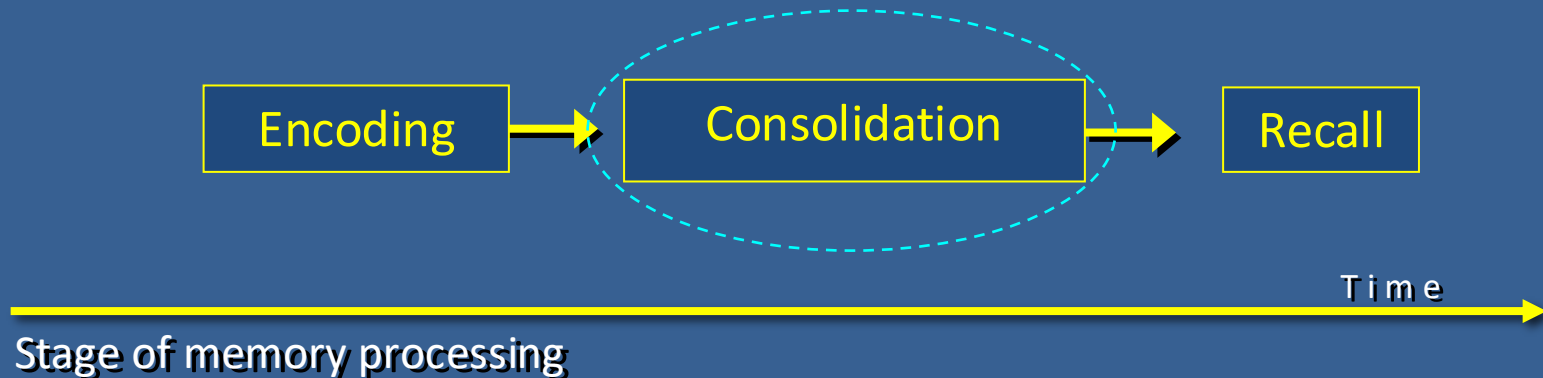


Need to simplify by posing specific questions.



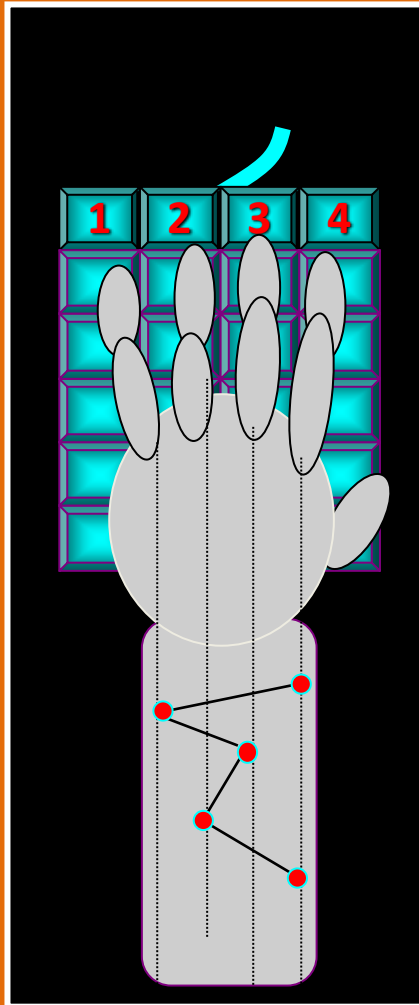
Memory Consolidation:

“A singular, time-dependent process involving the stabilization of learned information”



- Sleep & Memory I: The Question & Ingredients
- Sleep & Memory II: Procedural Skills - Motor

Finger-tapping Task:



“4-1-3-2-4”

Trial

Tapping 30s Rest 30s

Training

x12 Trials

Retest

x3 Trials

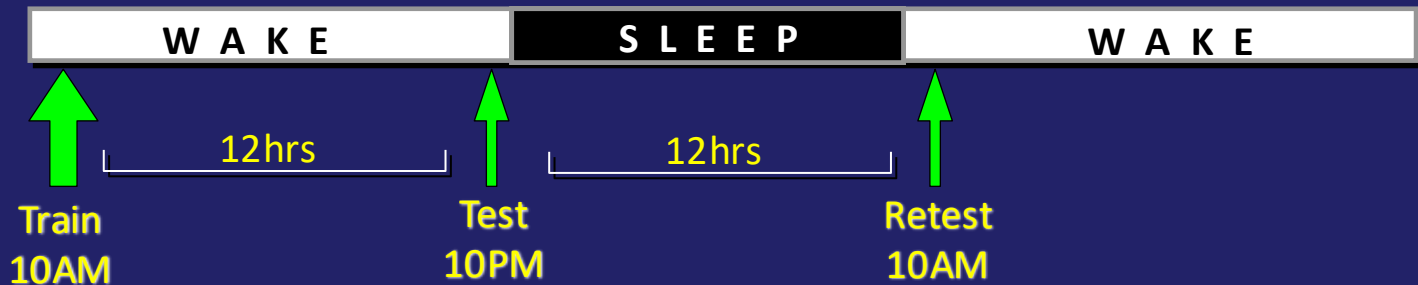
Outcome Measures

Speed = No. correct sequences per trial

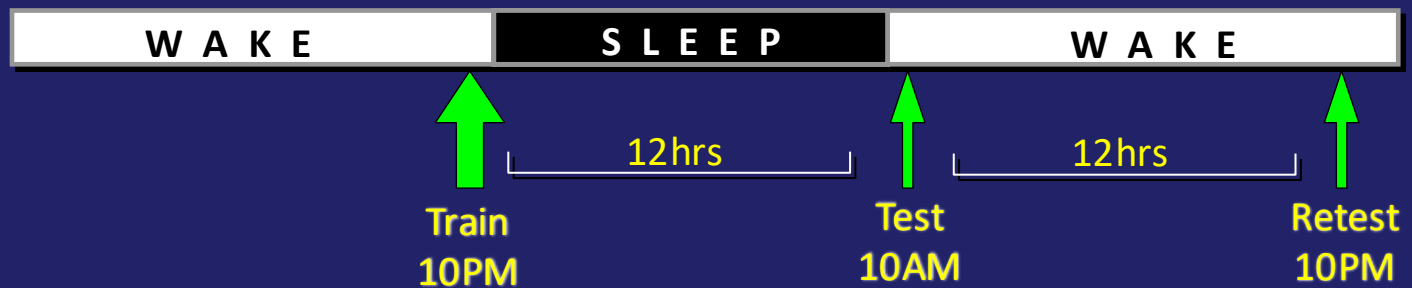
Accuracy = No. errors / sequence

Experimental Design

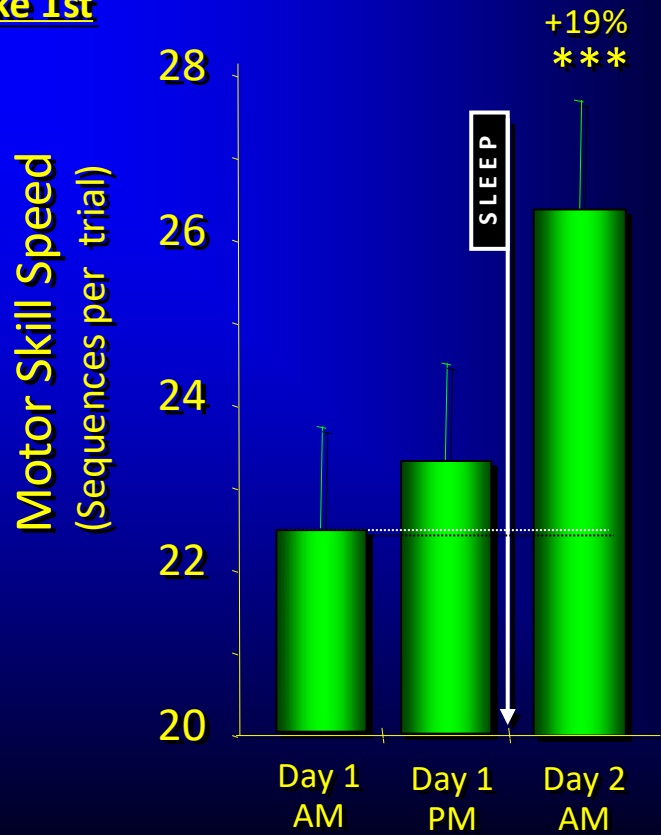
Wake 1st



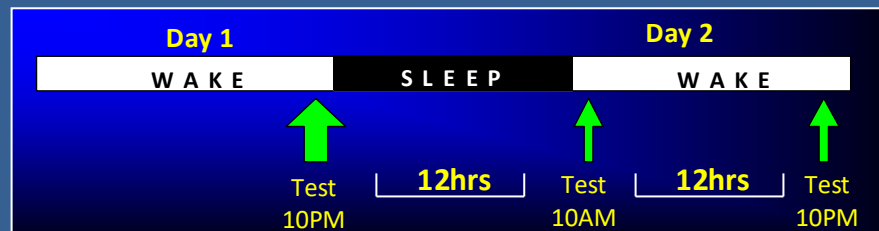
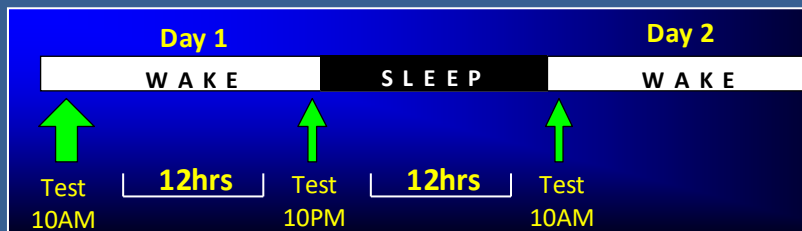
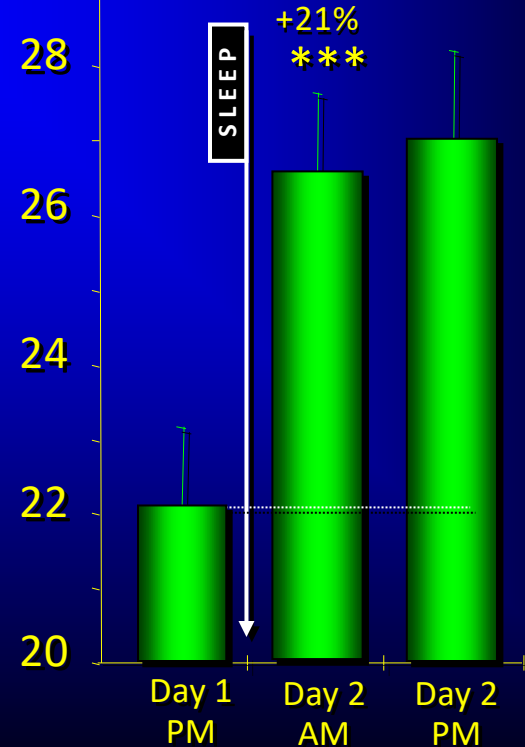
Sleep 1st



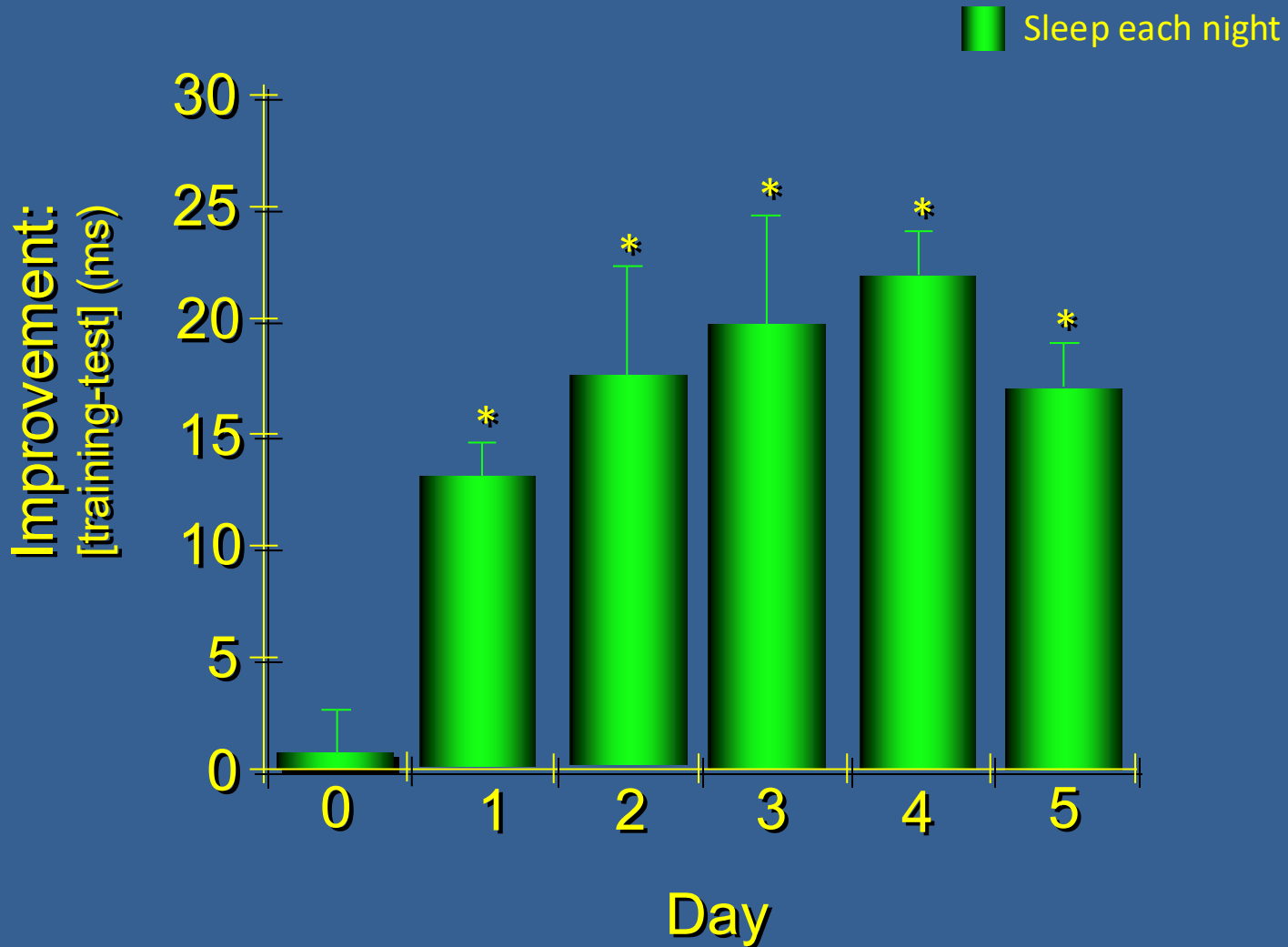
Wake 1st



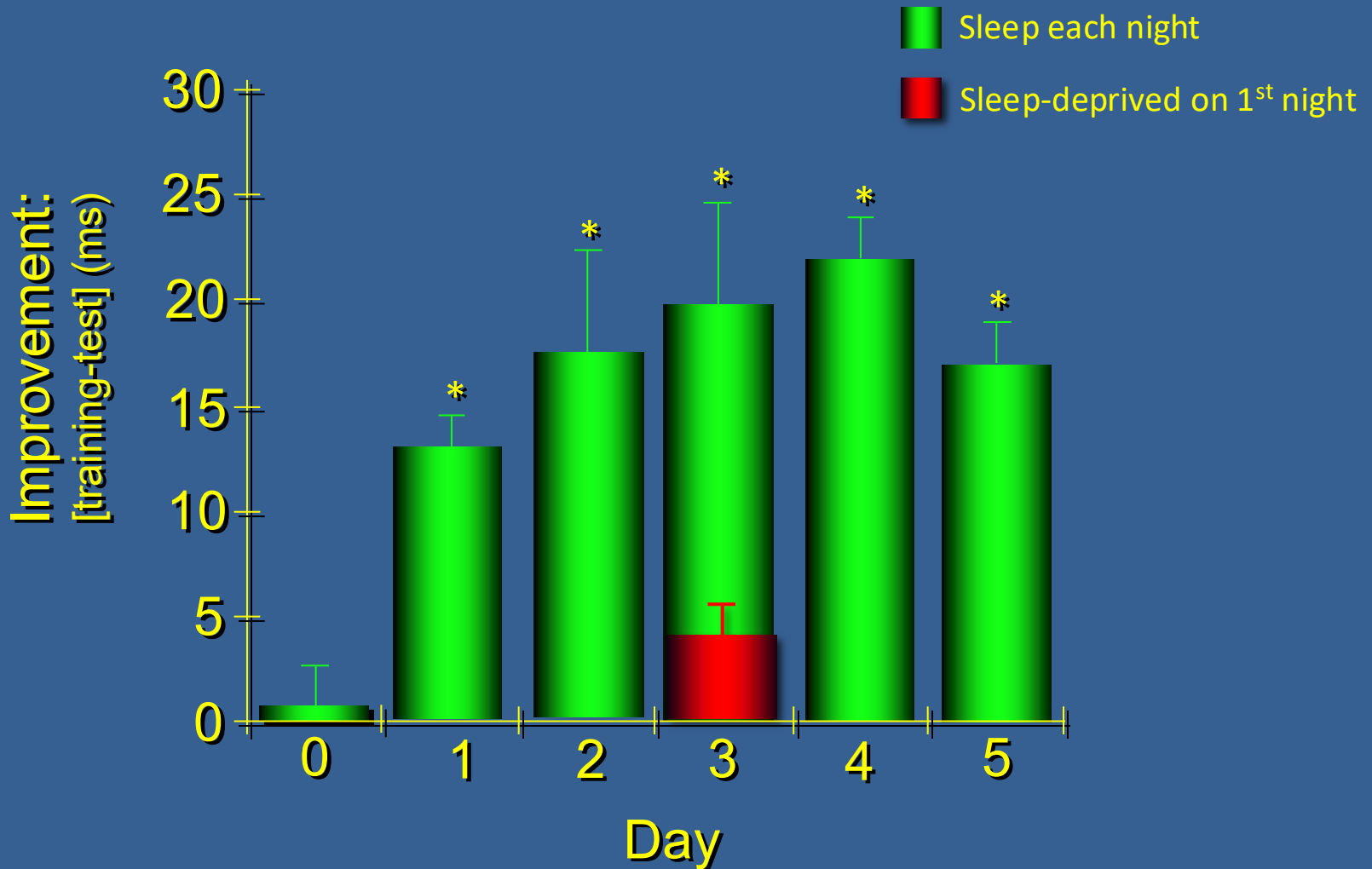
Sleep 1st



Sleep the night after the learning of a procedural task is important



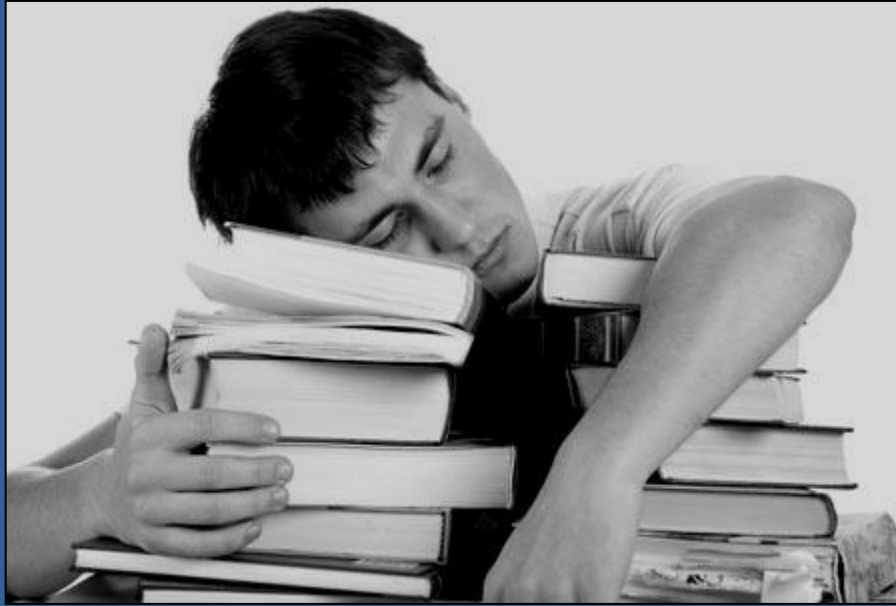
Sleep the night after the learning of a procedural task is important



Interim Summary

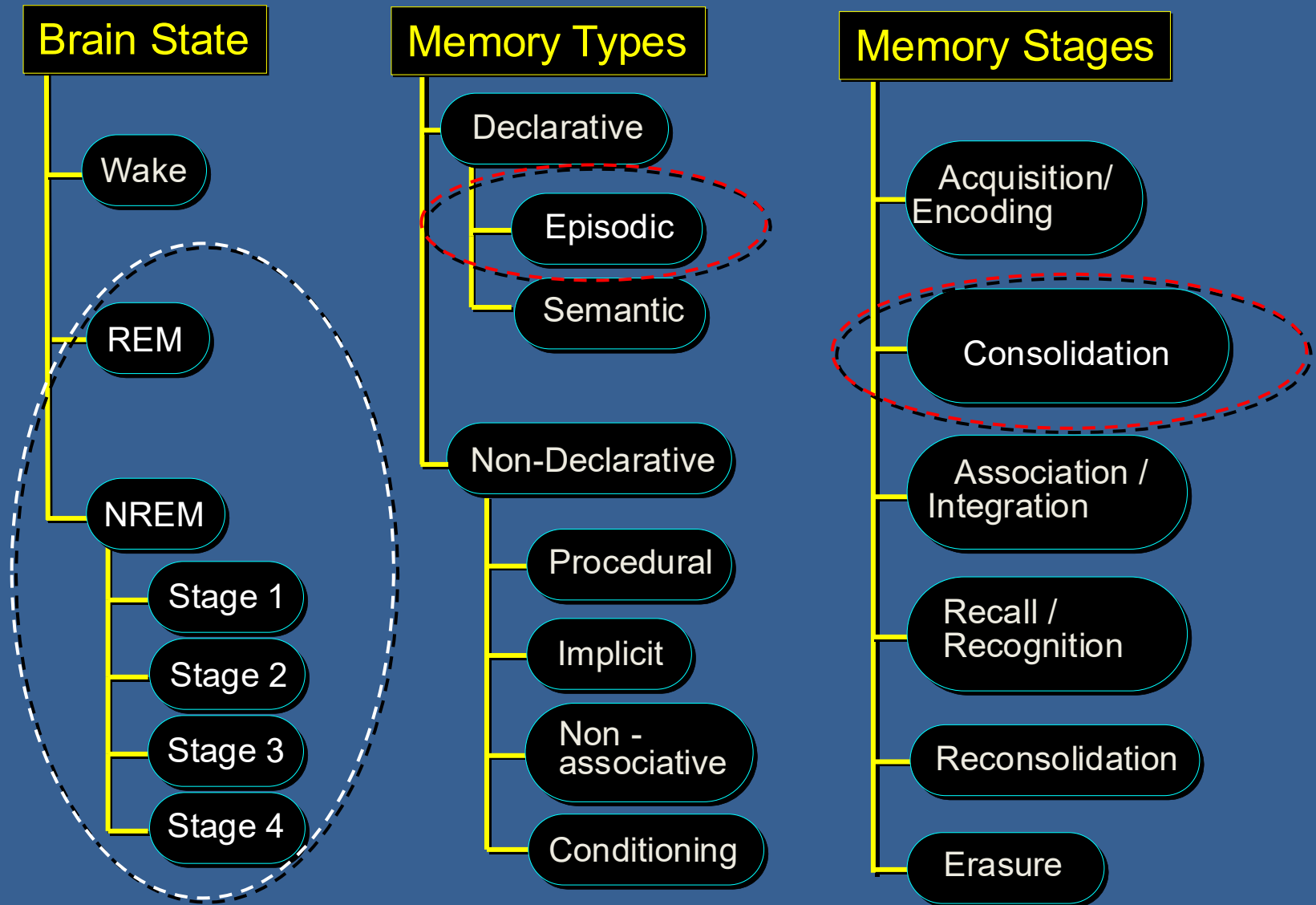
- The question of whether sleep effects memory is a complex blend of different types of sleep interacting with different types of memory, and different stages of memory
- For PROCEDURAL (“non-declarative”/skill) memories, sleep AFTER learning CONSOLIDATES and enhances memory.
- Sleep is not only ‘preferential’ for this step of consolidation, but absolutely necessary (1st night sleep deprivation results).

Sleep and Declarative Memory



- Sleep to retain facts
- Memory processing during sleep

Posing a new question:

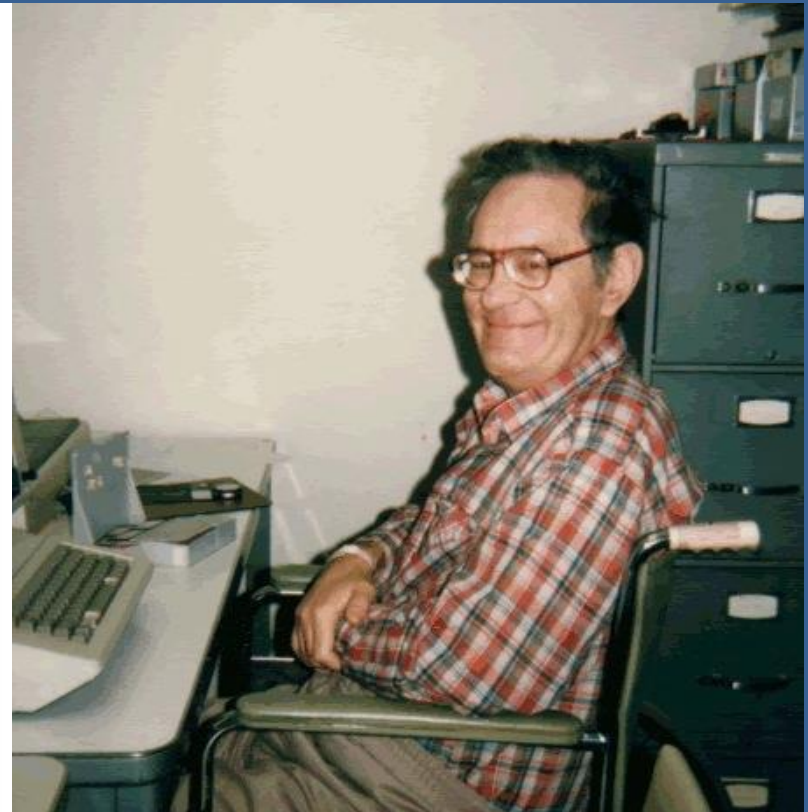
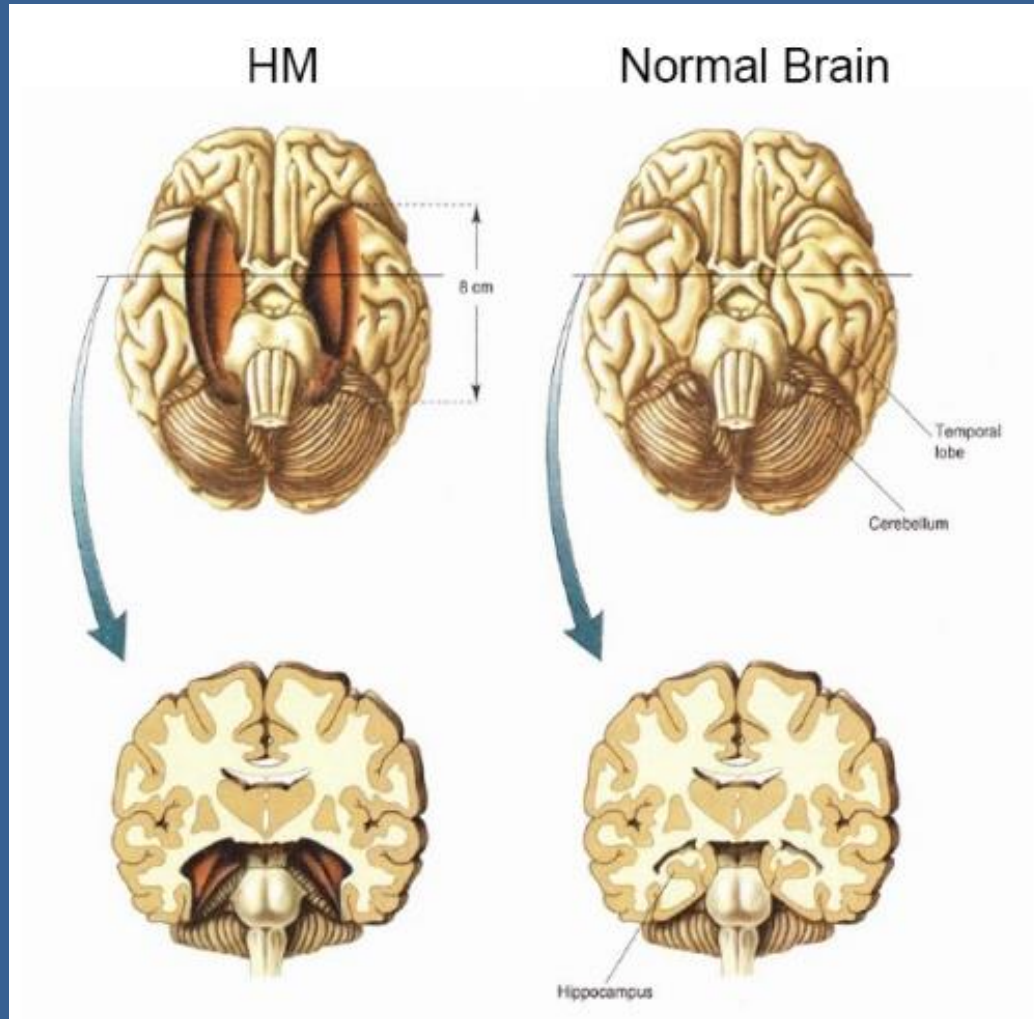


The hippocampus is important for episodic memory



Henry Molaison (HM)
1926-2008

The hippocampus is essential for declarative short-term and formation of long-term memory

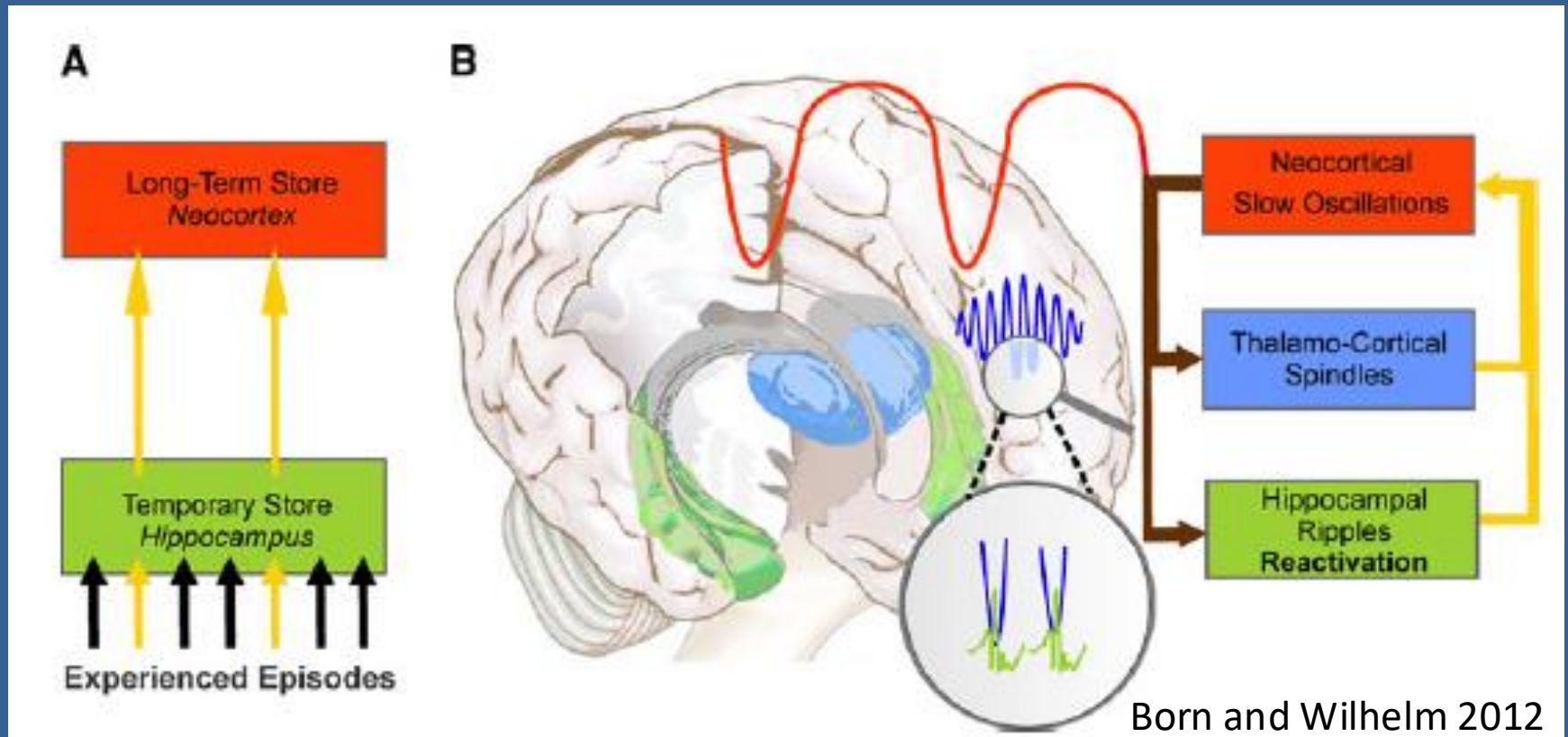


Specifically for declarative memory, not procedural memory. Declarative memories prior to surgery, maintained.

The Neurophysiology of Memory Consolidation

- The hippocampus is involved in the formation of episodic and spatial memory
- Both involve encoding and retrieval of temporally sequenced information
- The process of consolidation of these memories involve interactions between the hippocampus and the cortex during sleep
- Rhythmic brain activity organizes temporally sequenced information during its exchange between the hippocampus and the cortex.

A model of the fast and slow learning components of declarative memory consolidation.

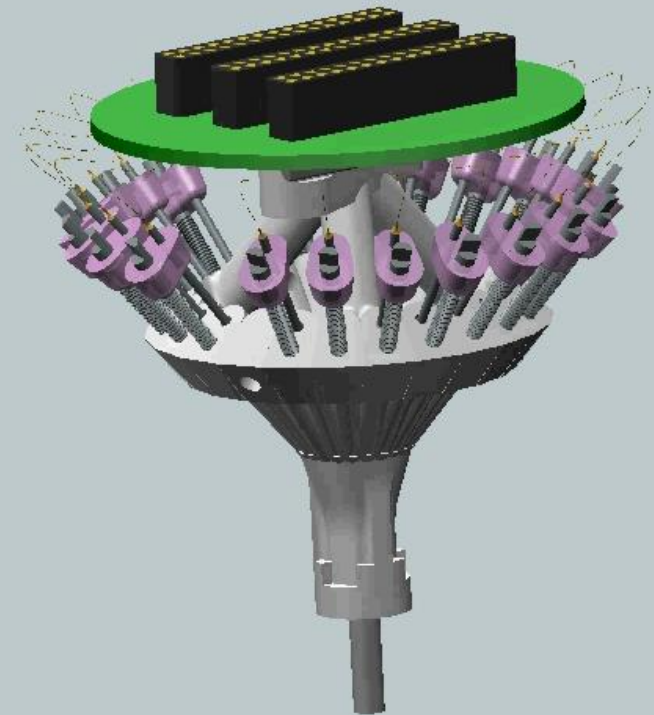


Predictions:

1. system consolidation process during sleep is selective, i.e. only some of the information encoded into the temporal store is transferred to the long term store whereas others become erased.
2. Memory can undergo specific qualitative changes during the off-line transfer from temporary to long-term storage.

Evidence for Encoding of Spatial Memory in Hippocampal Circuits

Microelectrode Microdrive Array for Chronic Recording ----- Wilson MIT

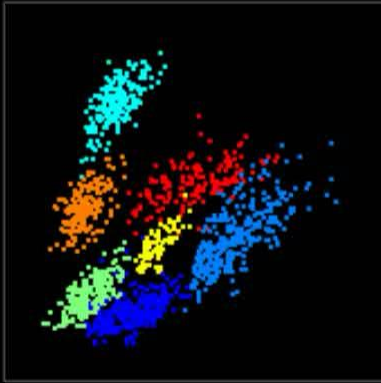


Allows simultaneous recording of 100 or more neurons at distributed sites around the brain in the behaving animal

Hippocampal place cell firing relates to spatial location of rat in maze.

cell activity

overall



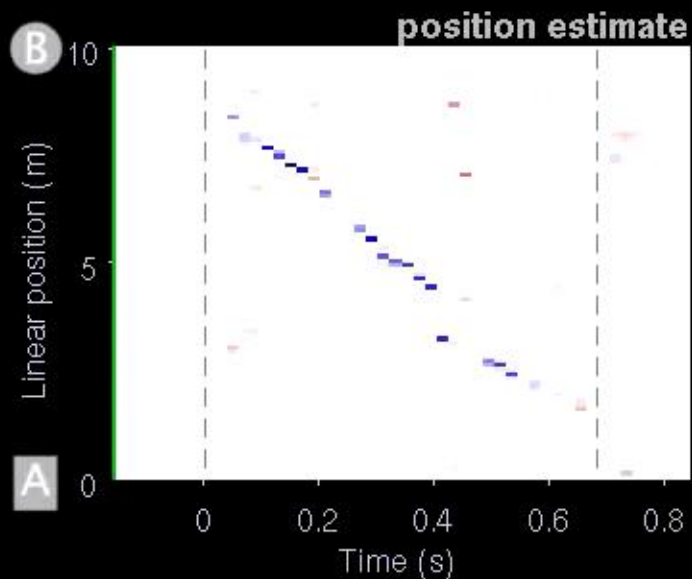
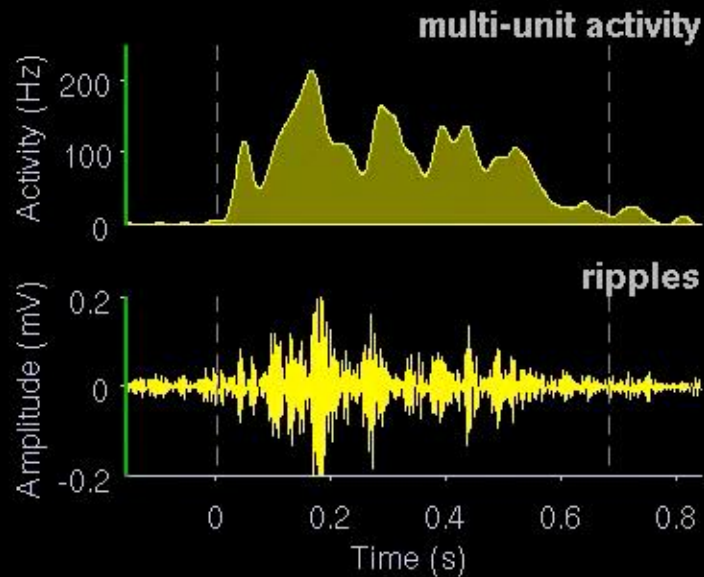
ongoing



behavior



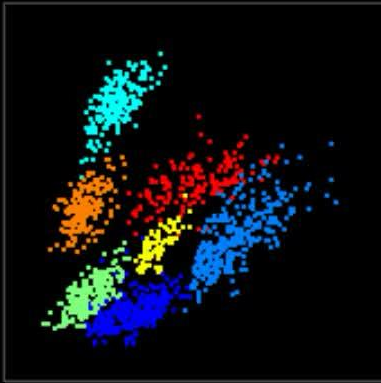
Reverse Replay from B to A



Reverse Replay During Sleep

cell activity

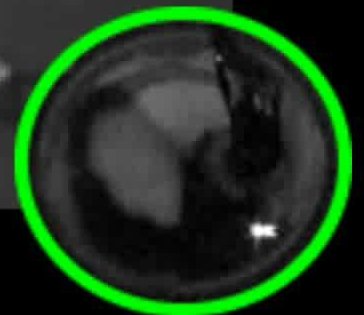
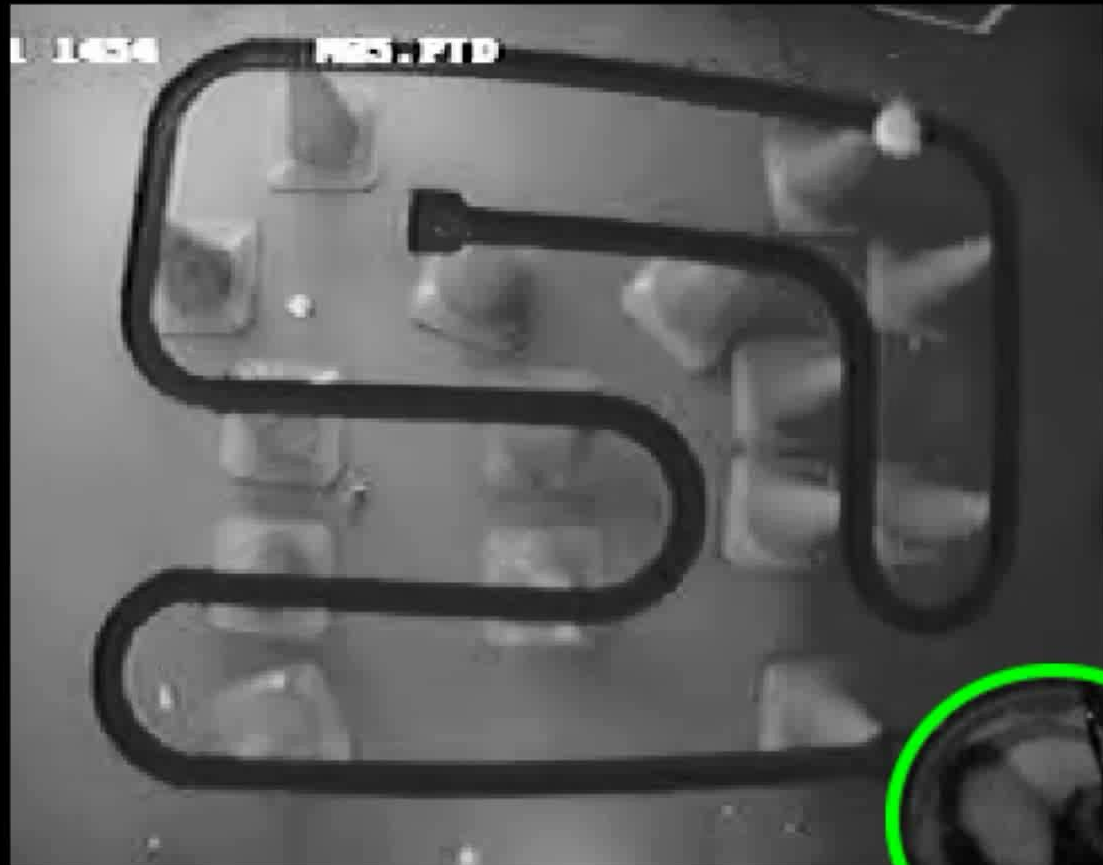
overall



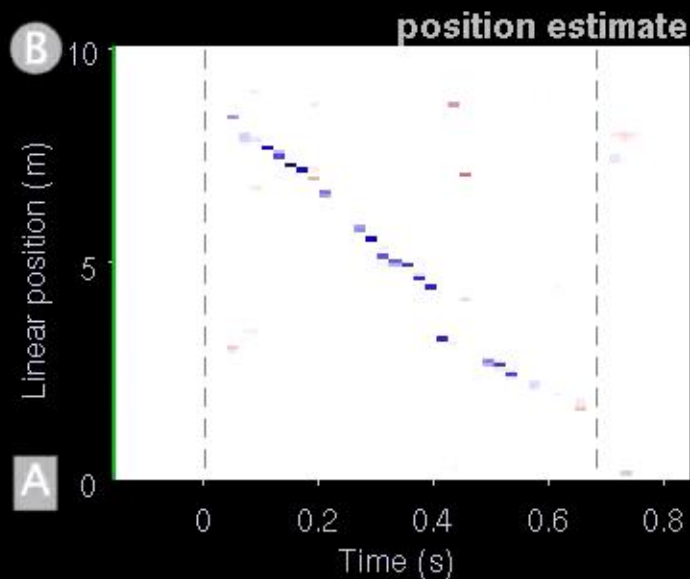
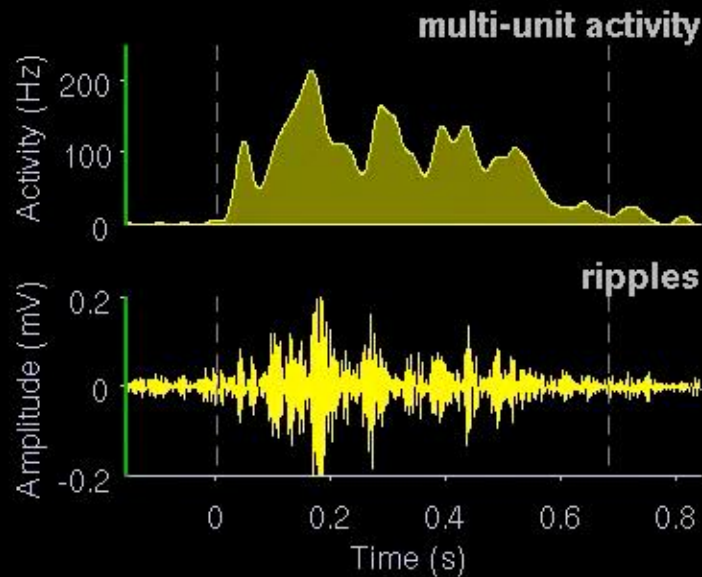
ongoing



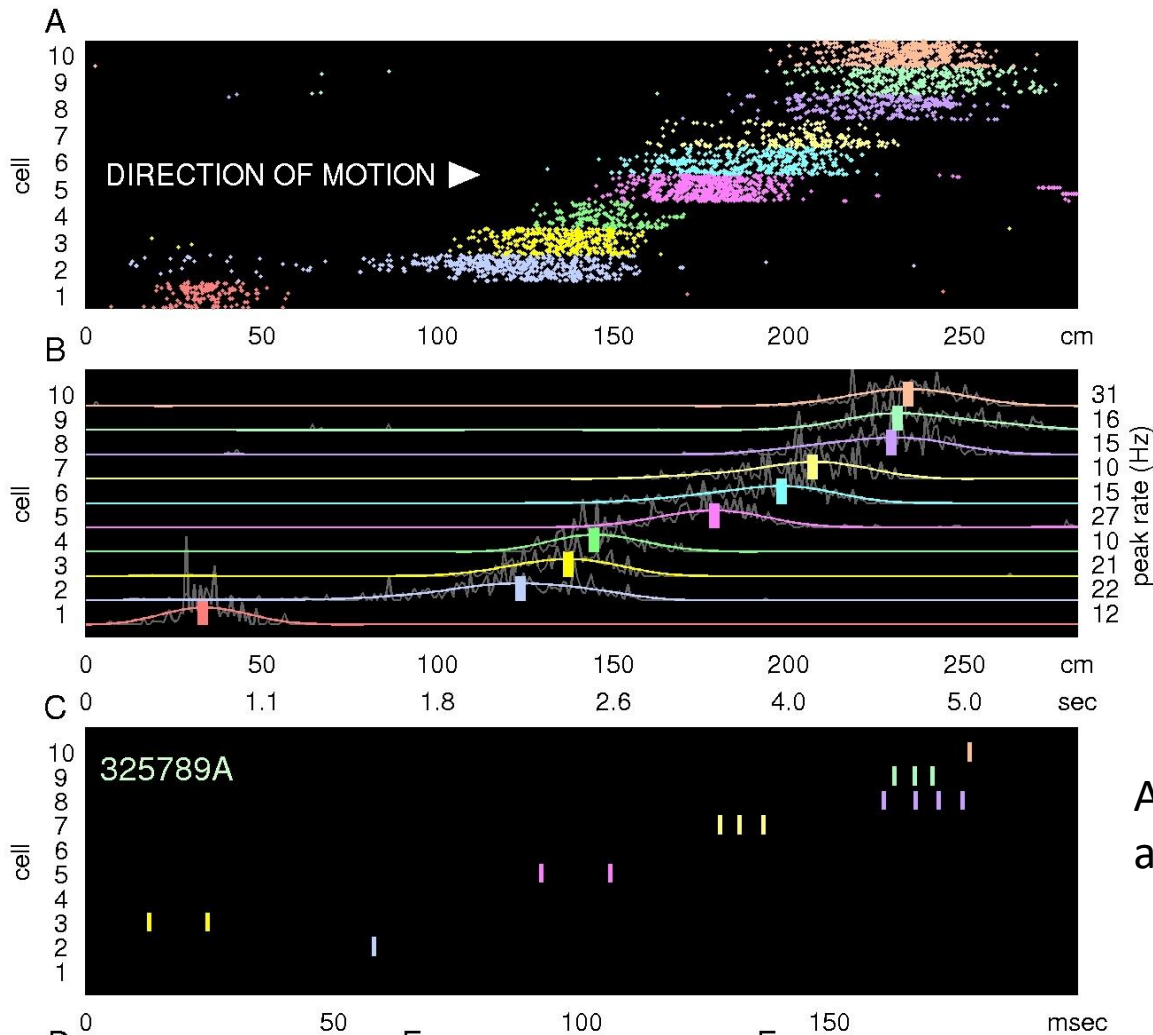
behavior



Reverse Replay from B to A



Firing of place cells in CA1 of hippocampus are replayed during sleep

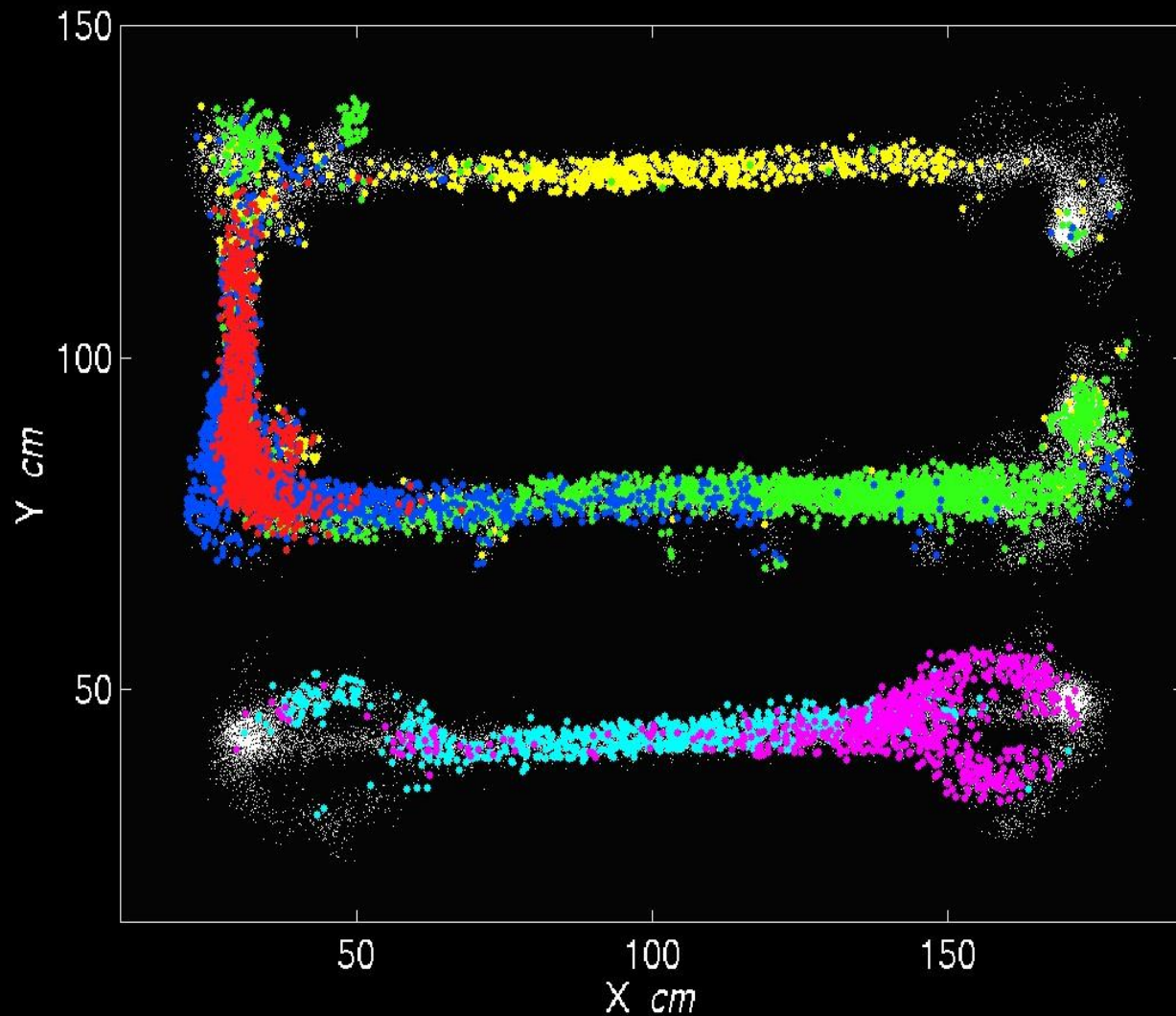


Forming Low Probability sequences when running maze

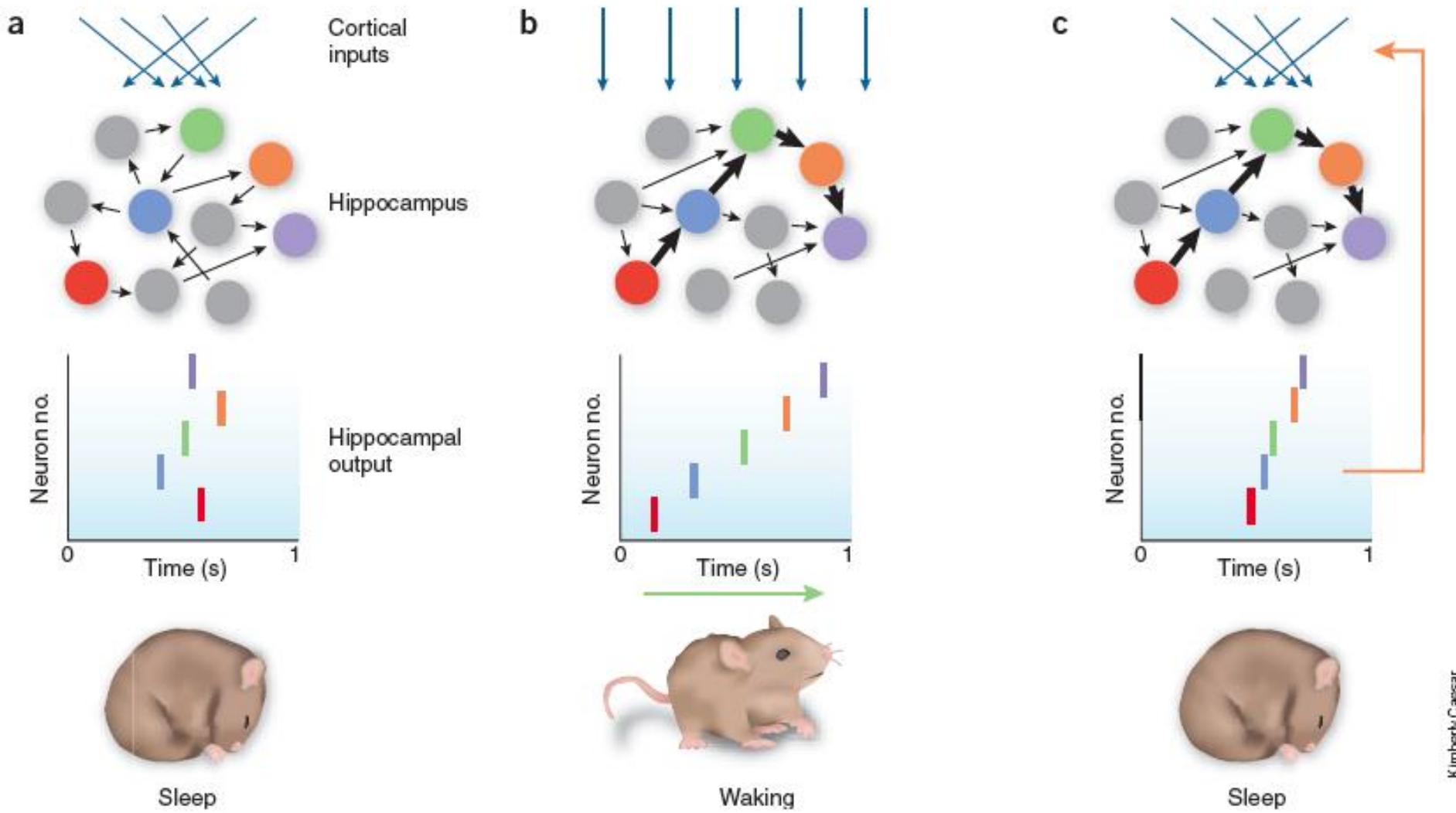
Averaged activity when running maze

Averaged Activity when sleeping after maze running.

Place Fields on Linear Tracks



Patterns of Firing of Hippocampal Place Cells Observed during Exploratory Behavior are Repeated During Subsequent NREM Sleep



Kimberly Caesar

What about in humans?

Can cued recall during sleep enhance memory?

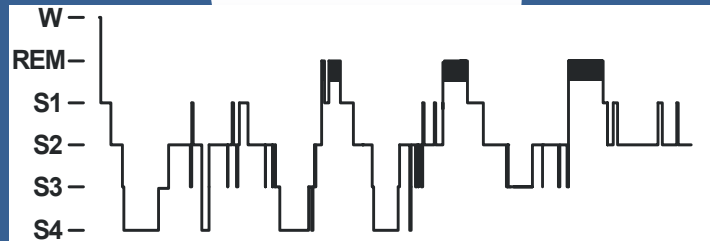
Learning



Consolidation



Retrieval



Odor reactivation during sleep! – Rasch et al. Science (2006)

The power of smell: “Retrieval Cue”

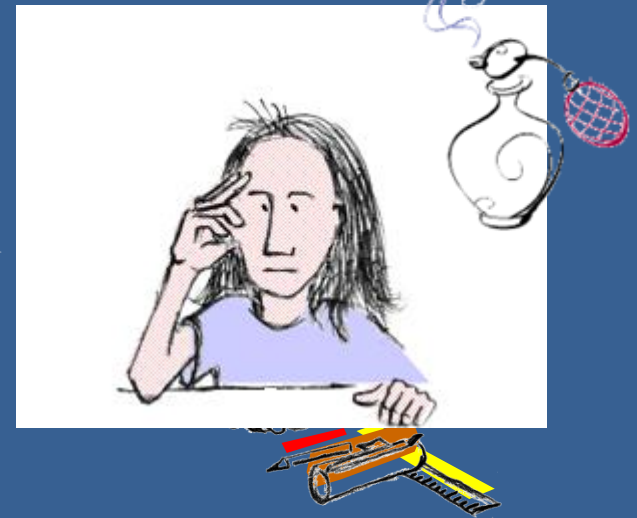
Learning



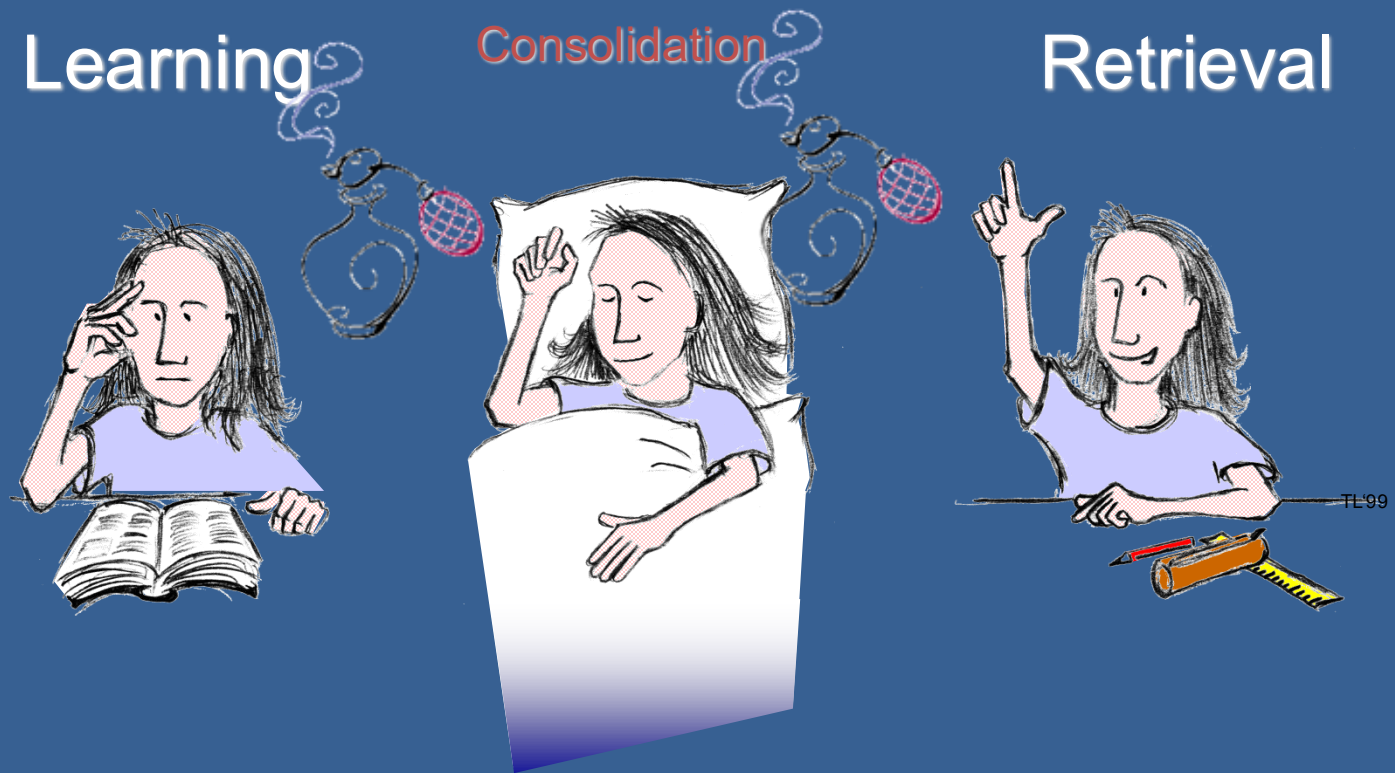
Consolidation



Retrieval

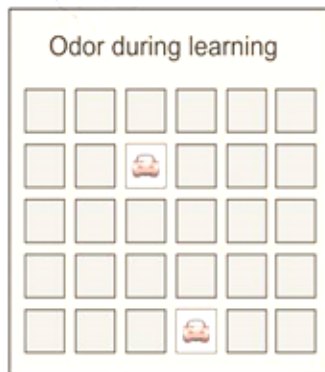


The power of smell: “Reactivation Cue”



Experimental Design:

Learning



20.00

24.00

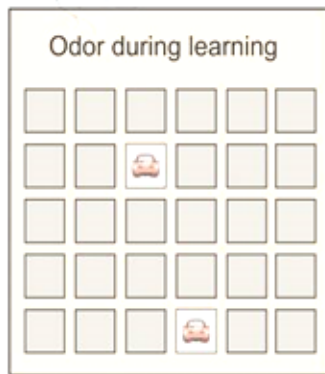
04.00

08.00

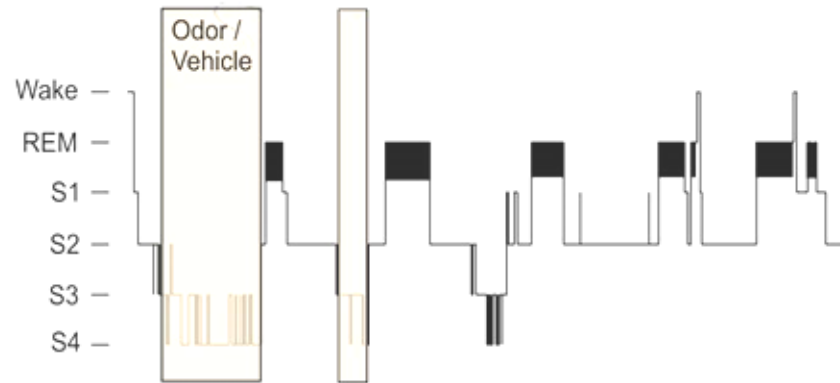
h

Experimental Design:

Learning



Sleep



20.00

24.00

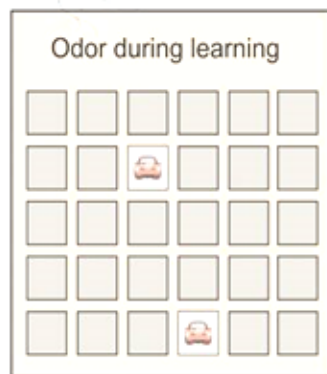
04.00

08.00

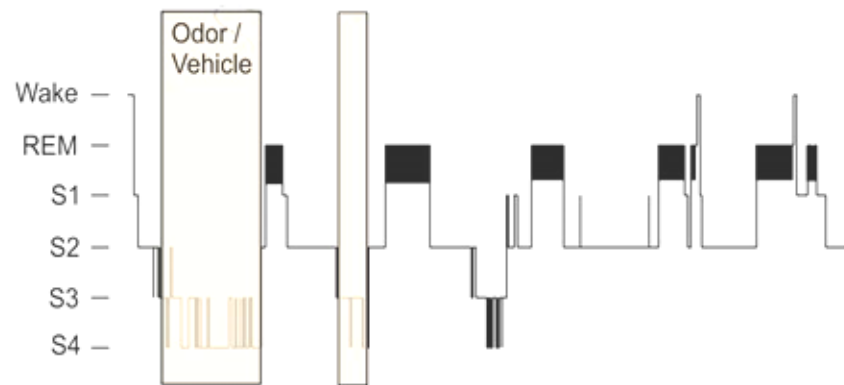
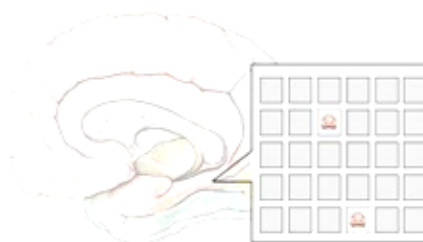
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Experimental Design:

Learning

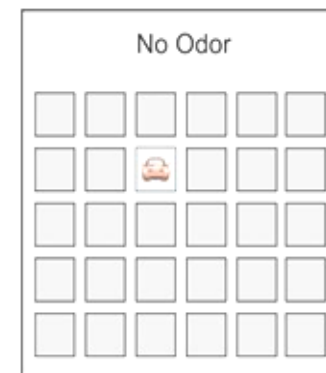


Sleep



Retrieval

?



20.00

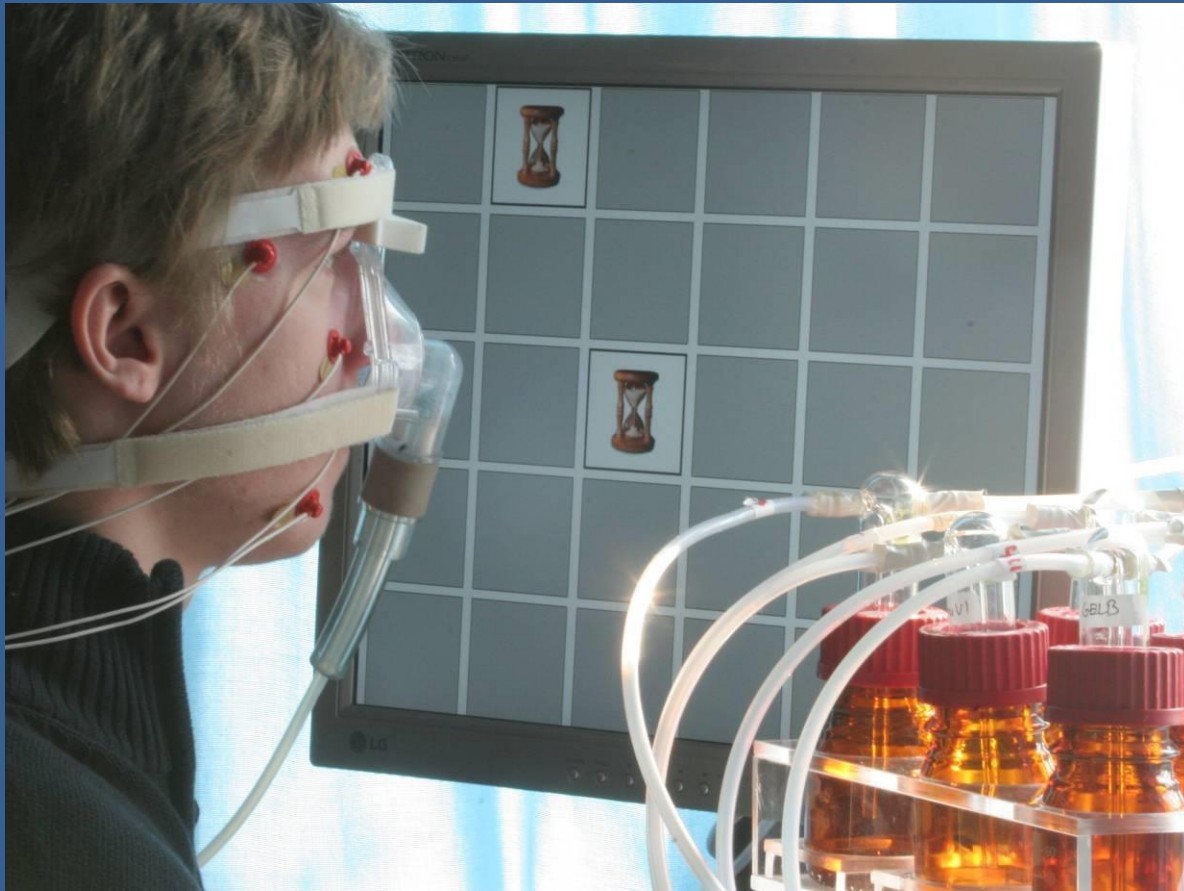
24.00

04.00

08.00

h

Learning...

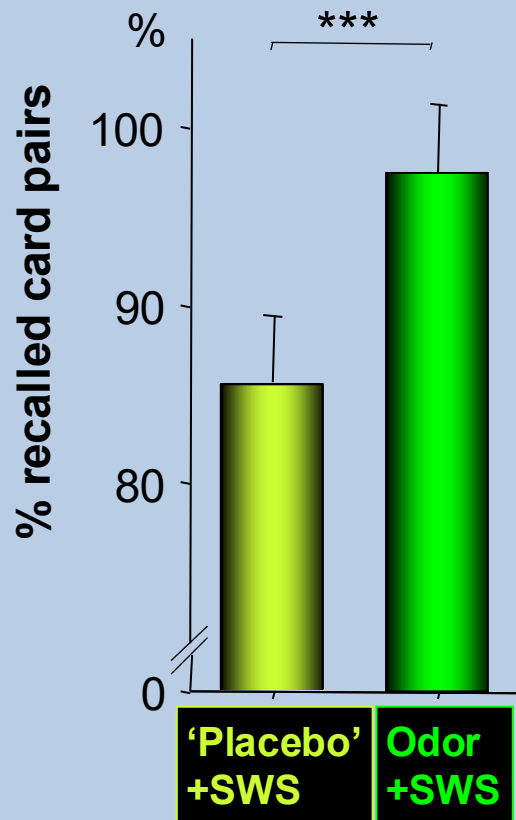


Sleeping...



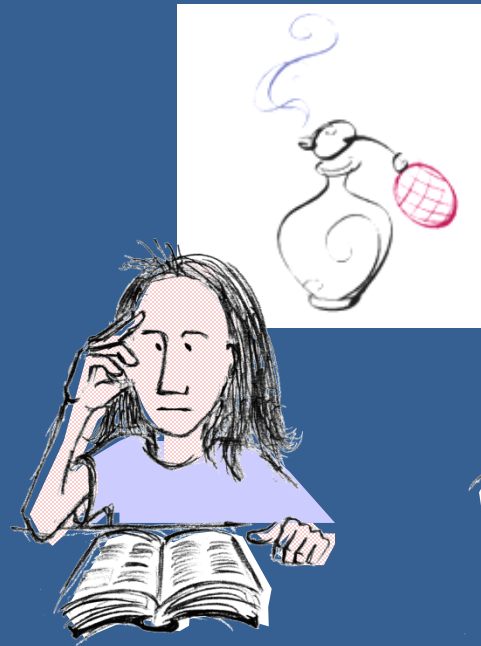
Results:

1. Memory for card pairs was enhanced after odor presentation during SWS.



Control experiment:

Learning



Consolidation

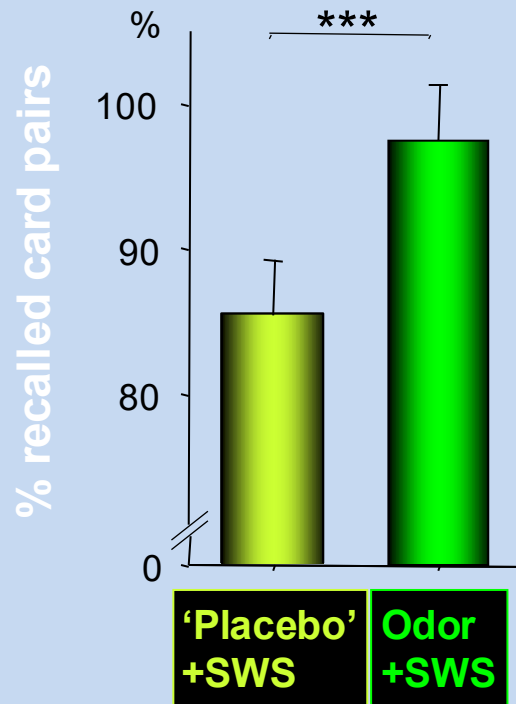


Retrieval

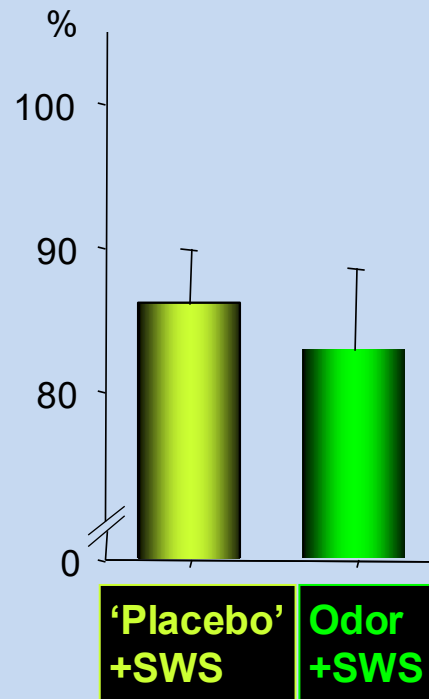


Results:

Main Expt



Control Expt



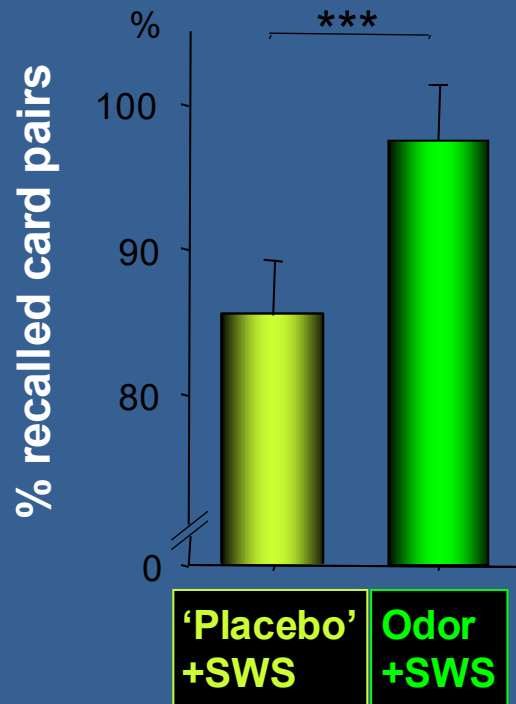
1. Memory for card pairs was enhanced after odor presentation during SWS.

2. The enhancing effect depends on the presence of the odor during learning.

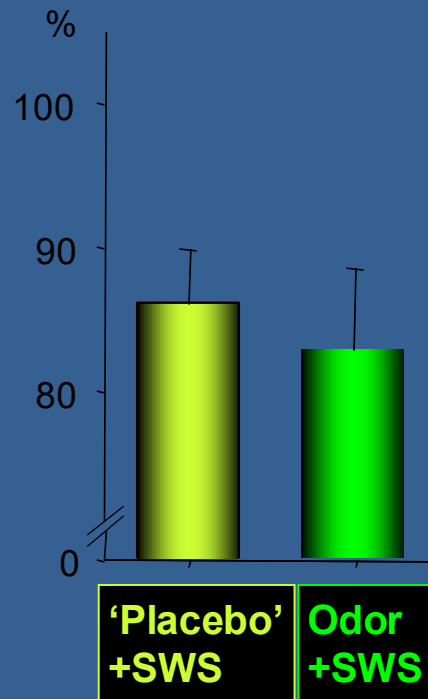
3. Odor during REM did not benefit memory, nor did odor during wake.

Results:

Main Expt



Control Expt



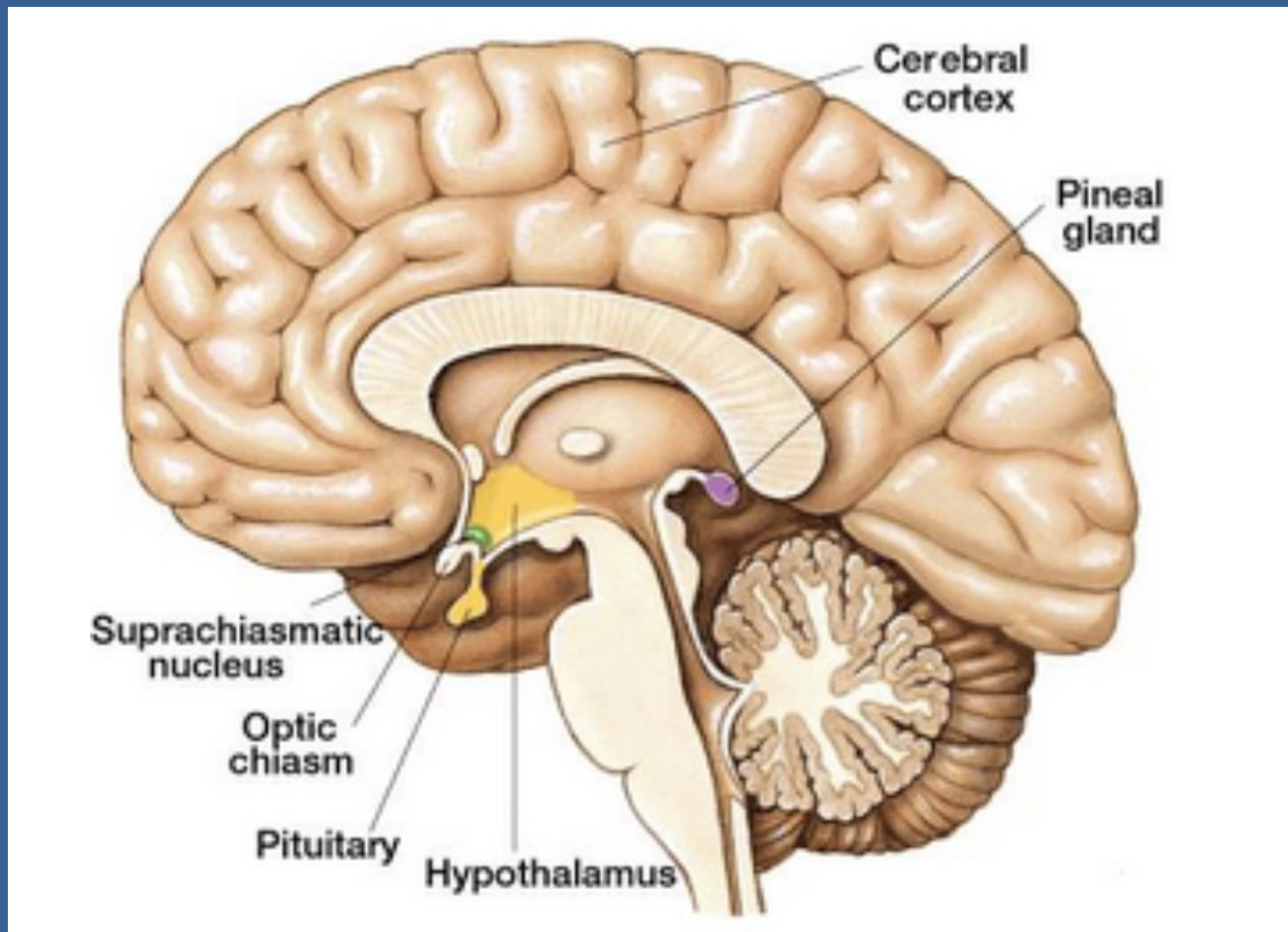
- Thus, sleep appears to be (re)processing / “reactivating” recently learned facts during SWS, thereby consolidating them.
- Moreover, this consolidation process can be amplified by experimentally “triggering” or cueing the reactivation of memories at night, based on the context [smell] associated with learning during the prior day

Interpretation: Reactivation of memory during sleep and subsequent reconsolidation enhances memory.



The role of circadian rhythms in learning and memory

The suprachiasmatic nucleus is the clock in the brain

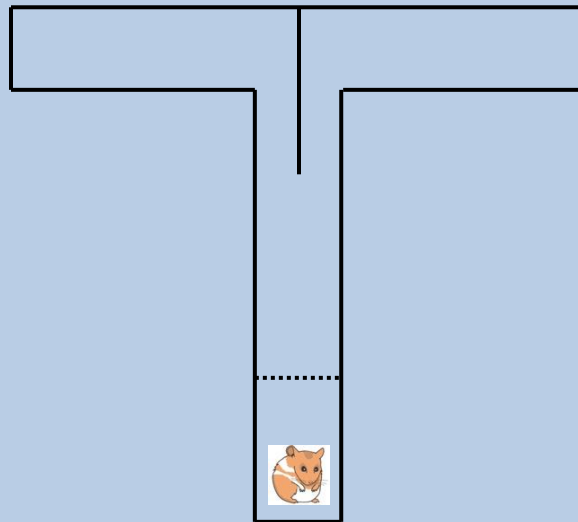


How can we investigate declarative memory stages and sleep/circadian influences in an animal model?

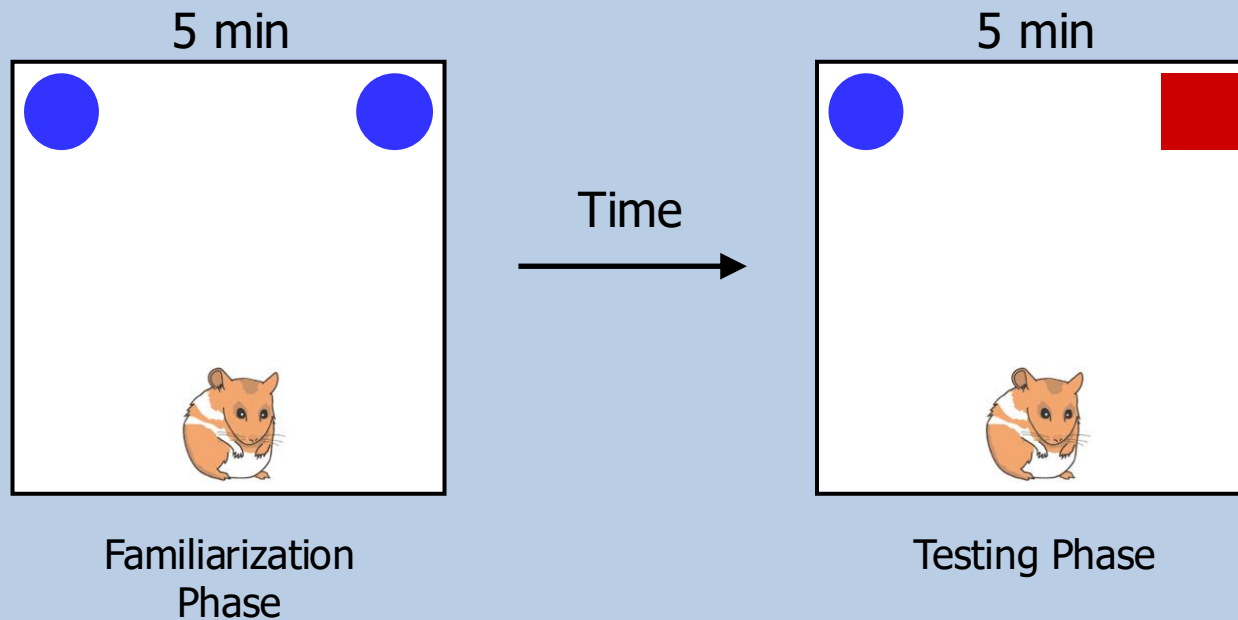


Meet the Siberian hamster

**Short-term or working memory is
assessed by spontaneous alternation
in a T-maze**

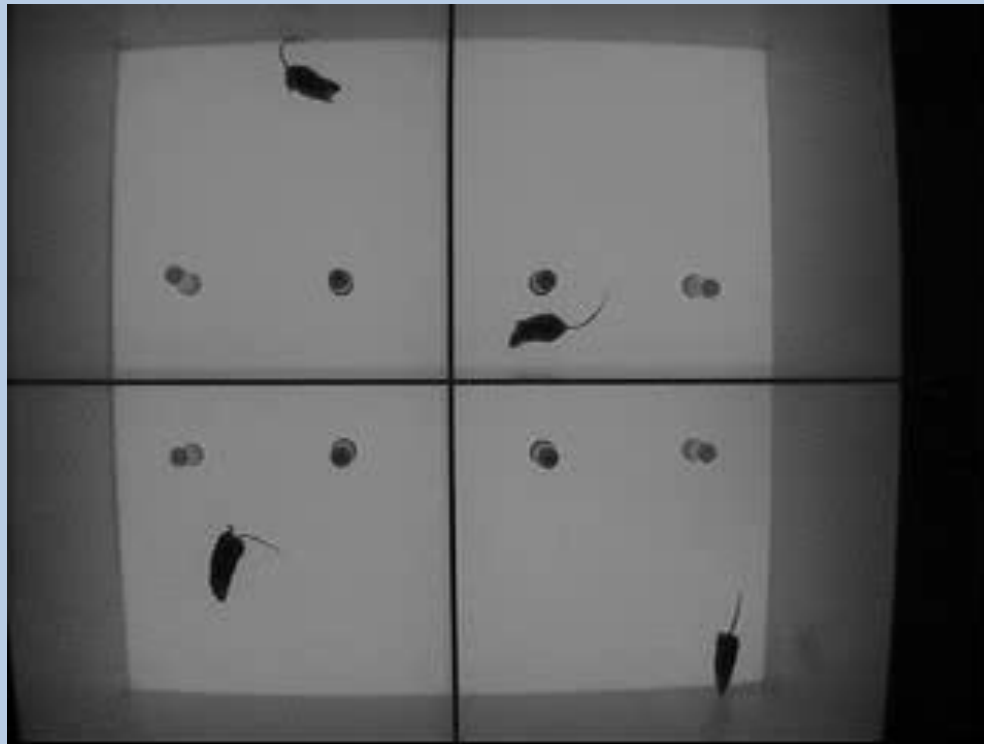


Long-term Learning and Memory is assessed by the novel object recognition and by novel object location test (NOR and NOL)



How do you assess the ability of mice to learn and form long-term declarative memories?

The Novel Object Recognition (NOR) Test Provides Excellent Biometric for Cognition

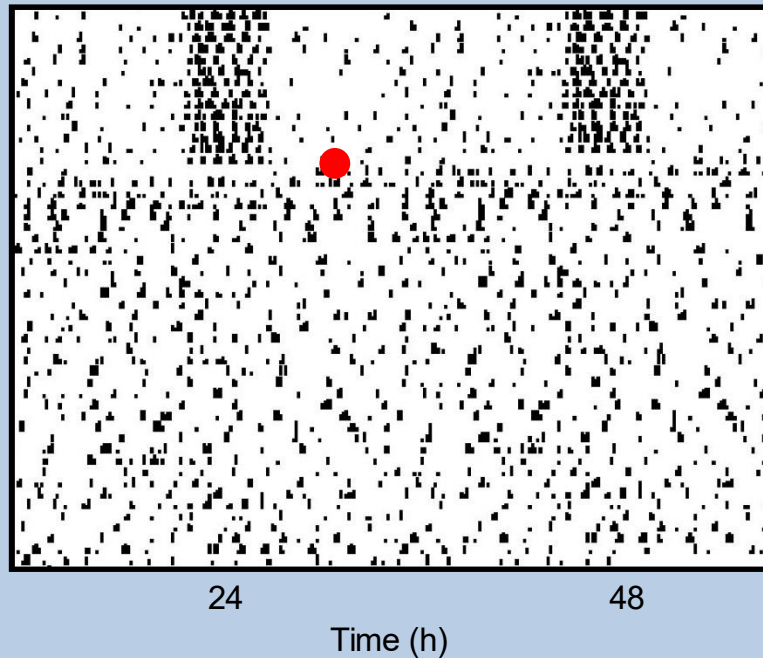
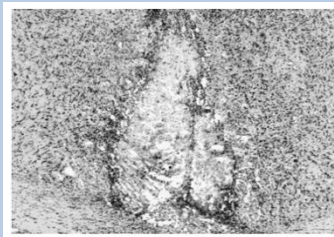
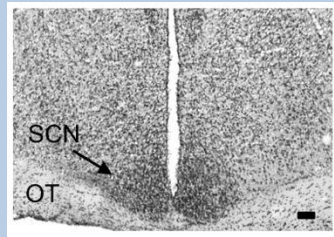


Other tests: open field, T-maze, hole board, fear conditioning, Barnes maze.

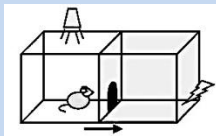
So, how can we distinguish between sleep and circadian effects on learning and memory?

You can't eliminate sleep, but you can eliminate circadian rhythms.

Suprachiasmatic nucleus (SCN) ablation model of arrhythmia



SCN Ablated Arrhythmic Rodents Perform Well in Memory Tests



Passive-avoidance
memory intact



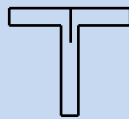
Recognition memory
intact



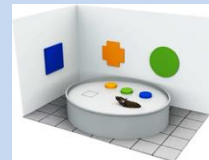
Recognition memory
intact



Time place memory
intact



Spatial memory
intact



Water maze learning
intact

Stephan and Kovacevic,
Behav Biol, 1978

Ruby et al., *Science*,
2014

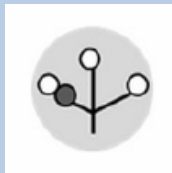
Phan et al., *J Neurosci*,
2011

Mulder et al.,
Chronobiol Int, 2013

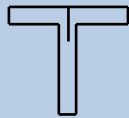
Cain et al., *Behav Brain
Res*, 2012

Clock gene knockout models of arrhythmia

per1^{-/-}/per2^{-/-}



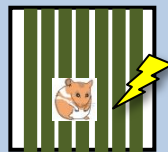
Time place
memory intact



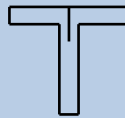
Spatial memory intact

Mulder et al., *J Biol Rhythms*, 2013

cry1^{-/-}/cry2^{-/-}



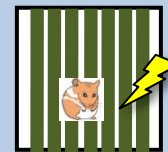
Contextual fear
memory intact



Spatial memory intact

van der Zee et al., *Curr Biol*, 2011

bmal1^{-/-}



Contextual fear
memory intact



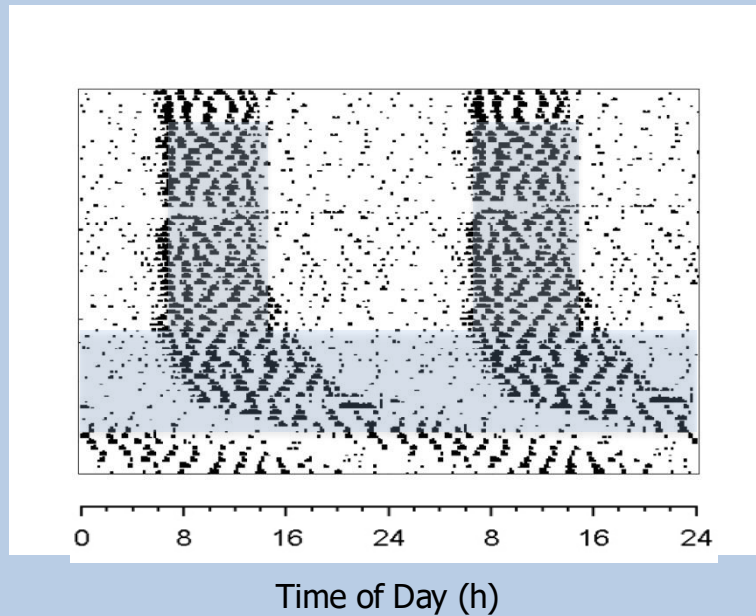
Recognition memory intact

Wardlaw et al., *Learning Memory*, 2014

Also, no deficits seen in learning and memory tests.

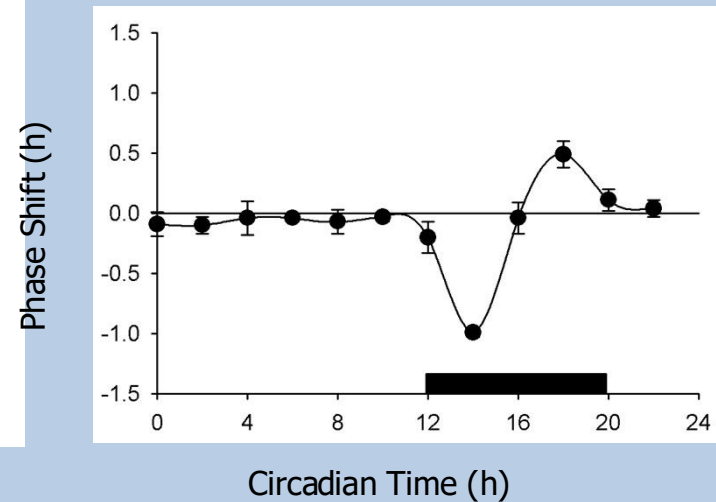
Siberian Hamsters are Model Circadian Animals

They freerun and follow
Aschoff's Rule

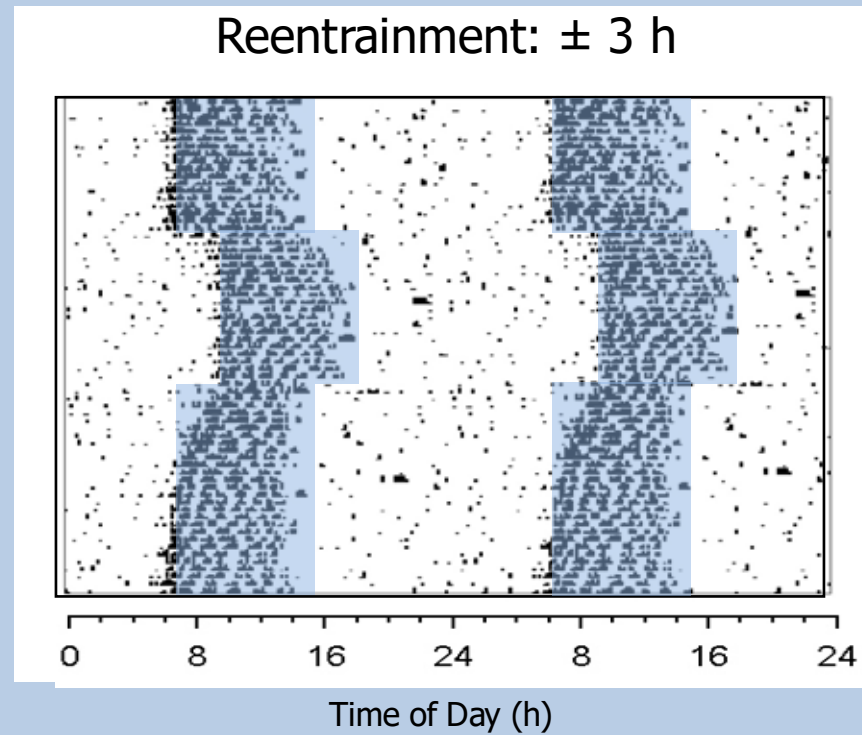


They have normal
phase response curves

Light pulse early night phase delays



They re-entrain to short shifts in the photocycle:

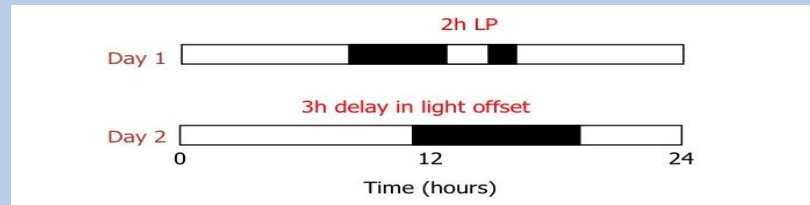


Ruby et al.

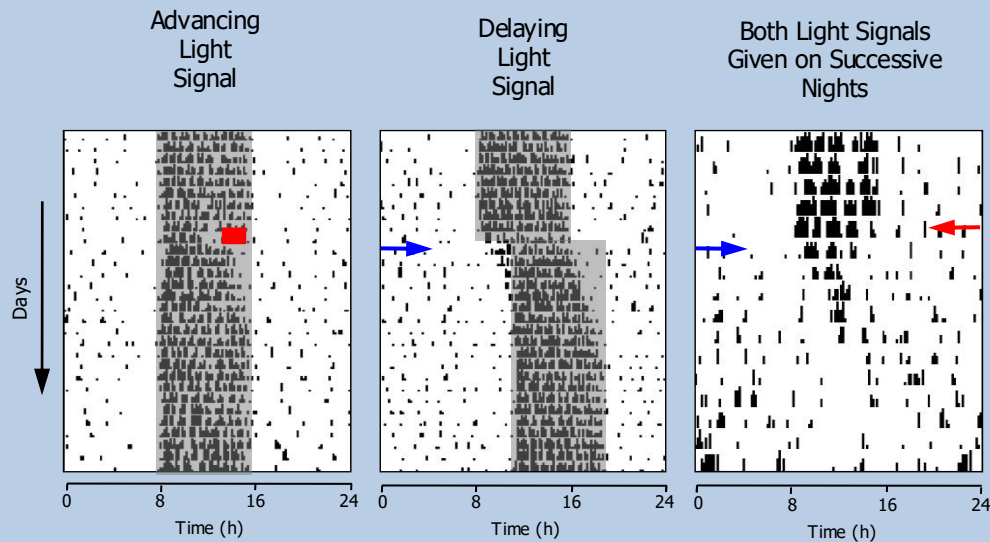
But, Opposing light signals induce arrhythmia

a phase advance

a phase delay



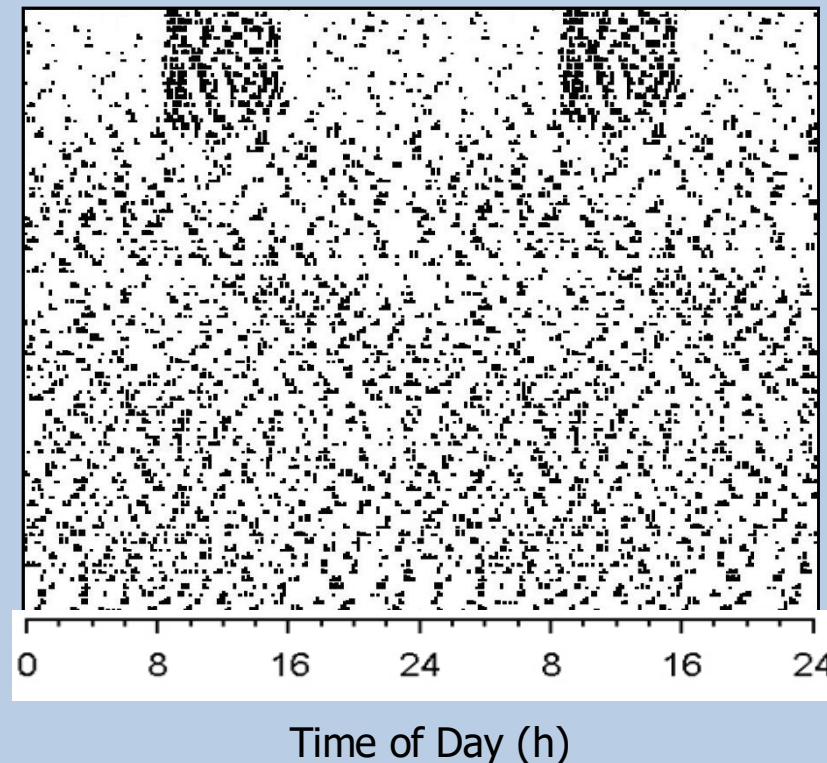
Followed by



THE DISRUPTIVE PHASE SHIFT (DPS) PROTOCOL

Fernandez, Ruby, *et al.*, *Science*, 2014

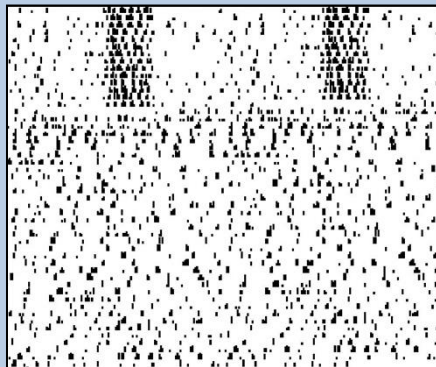
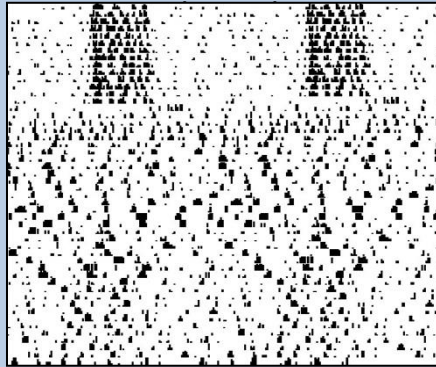
In Siberian Hamsters, DPS Results in Loss of Circadian Rhythmsfor Life.



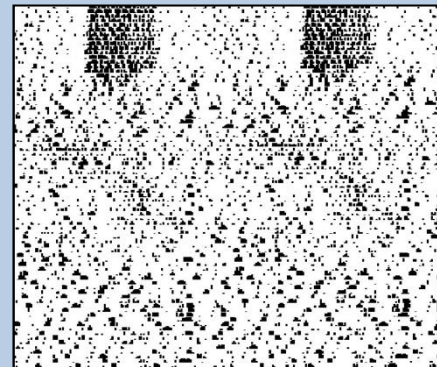
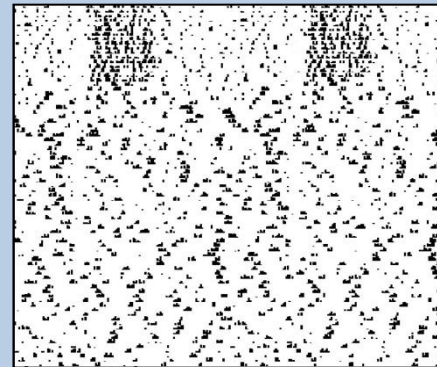
A small percentage of animals re-entrain.

Does arrhythmia per se impair learning and memory in Siberian hamsters?

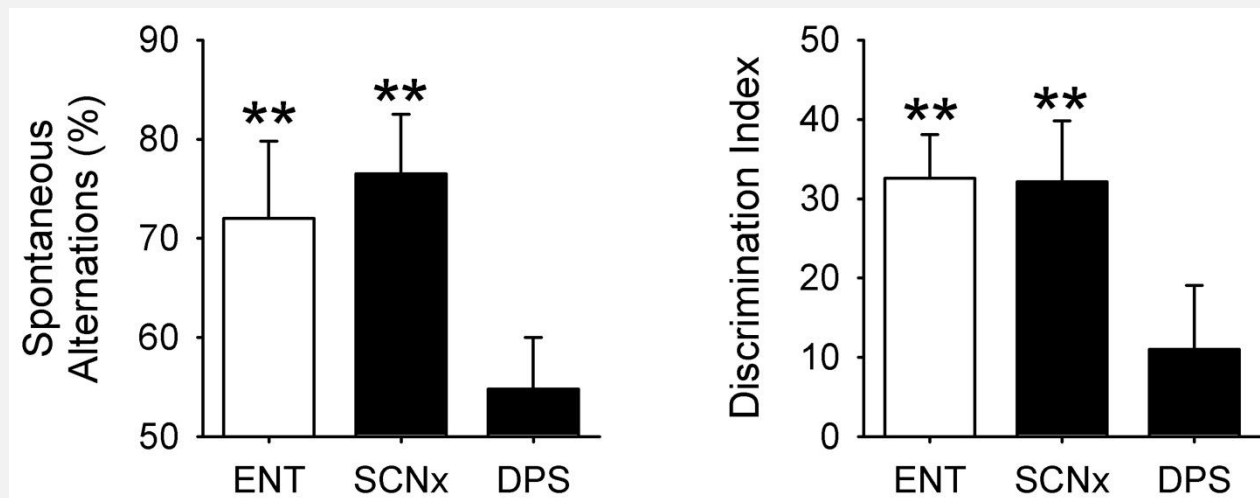
SCN Lesion



Disruptive Phase Shift (DPS)

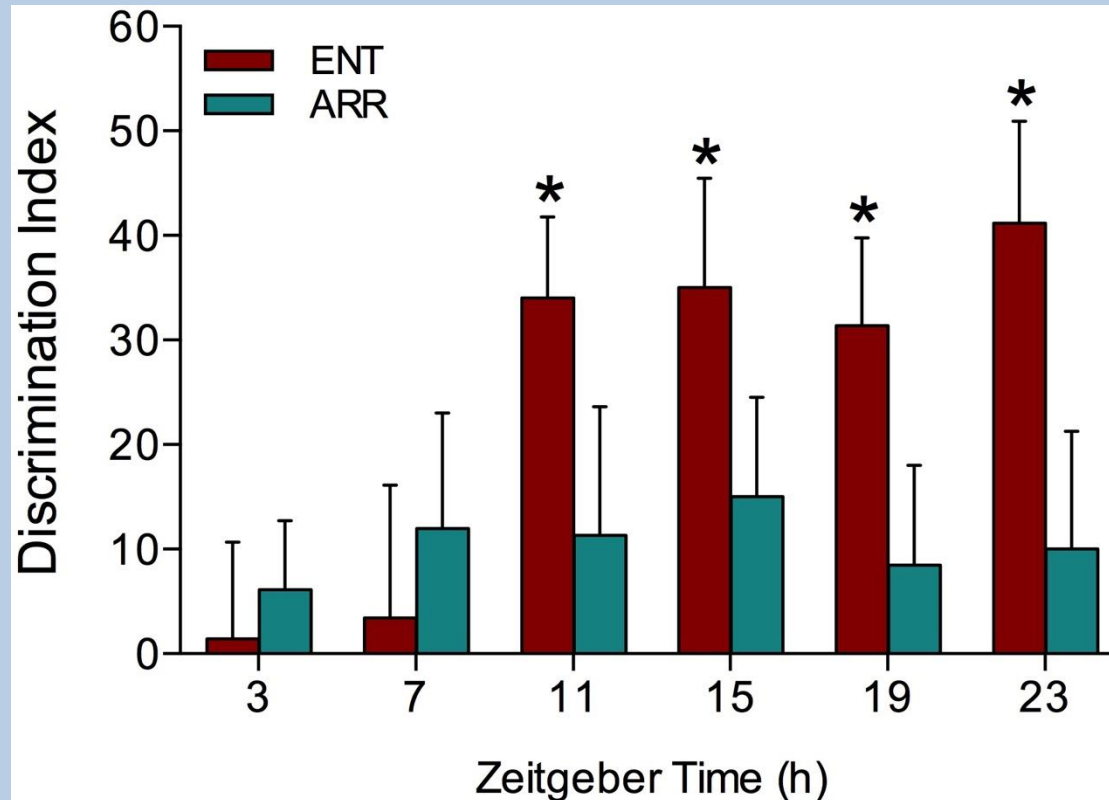


In the Siberian hamster as in other rodents, SCN ablation has no effect on SA or NOR, but the DPS does.

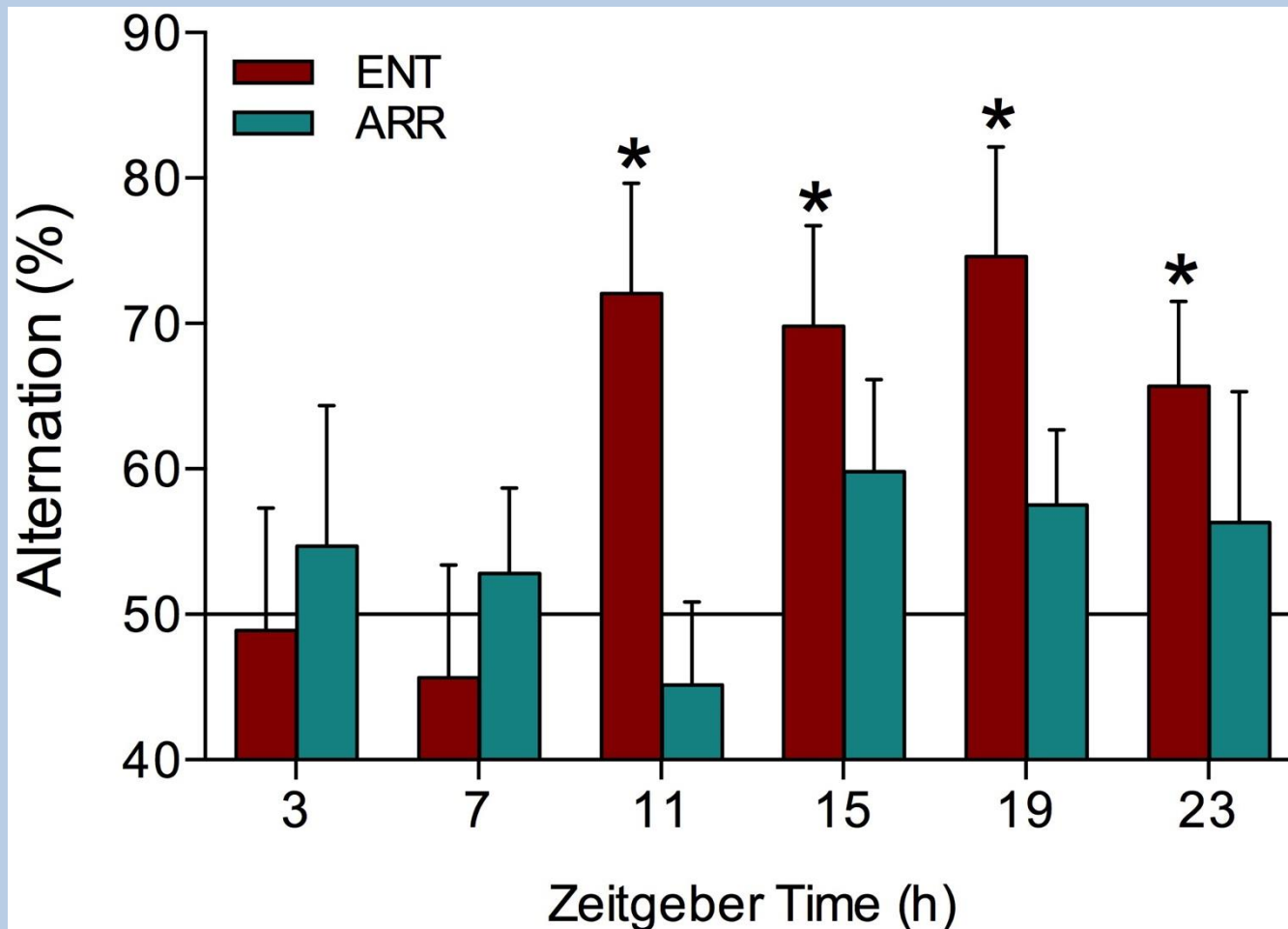


The difference is the presence of a functional but desynchronized SCN in the DPS animals

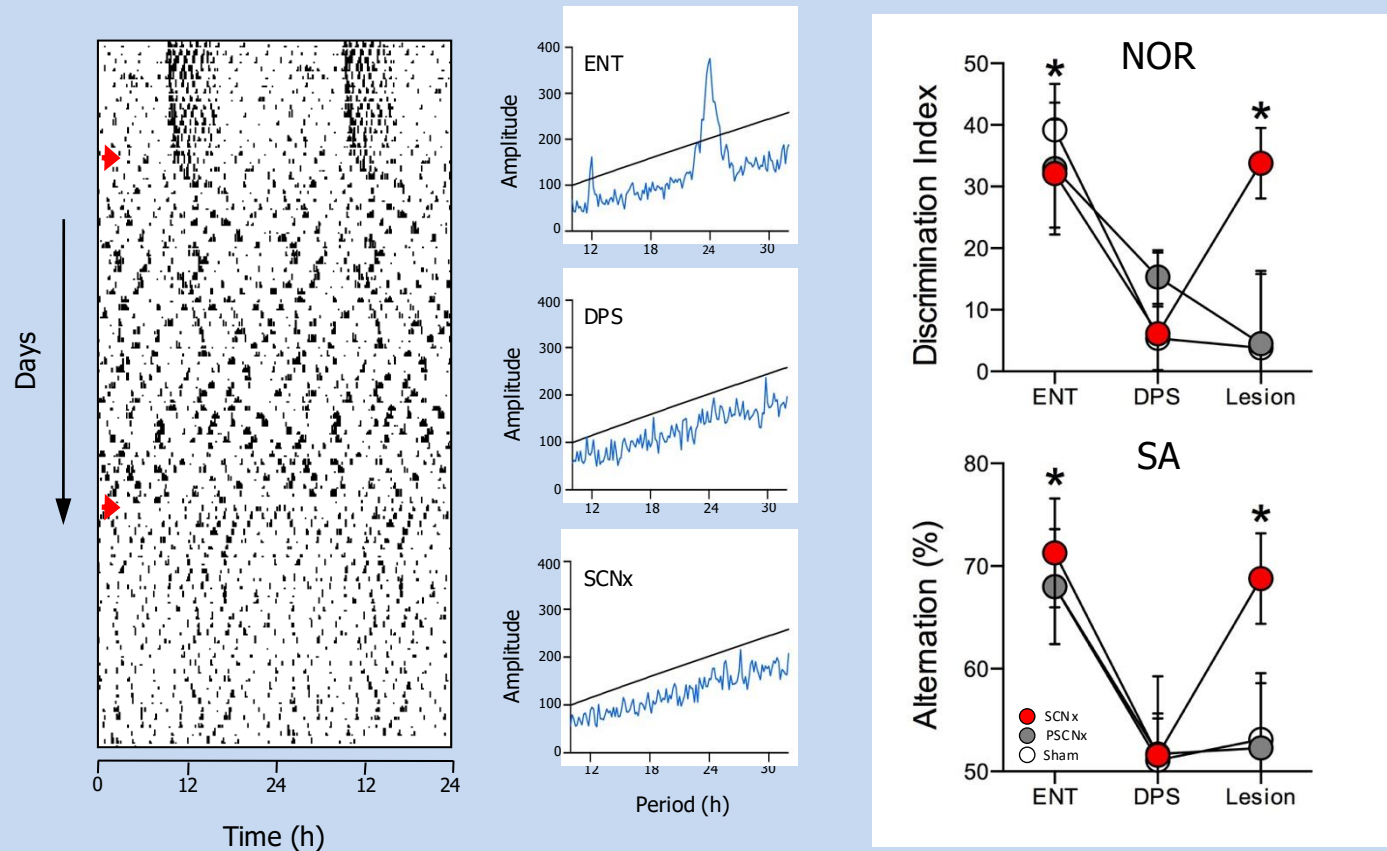
Novel object recognition (interval=60 min)



Spontaneous Alternation



Is the presence of an arrhythmic SCN responsible for learning deficit? Can memory be rescued by SCN lesion?



Yes, SCN ablation restores learning and memory abilities in the arrhythmic hamster.

Possible explanation for SCN inhibition of neuroplasticity:

- Short-term declarative memories formed during wake.
- Those memories are consolidated during sleep.
- Consolidation is transfer from hippocampus to cortex.
- SCN inhibition of neural plasticity protects the fidelity of the transferred information.

A photograph of a man with a beard, wearing a light-colored long-sleeved shirt and dark pants, sleeping peacefully under the chassis of a green vehicle. He is lying on his side, with his head resting against the underside of the vehicle and his hand near a large, worn tire. The vehicle has a yellow warning light on top. The background is a dark, textured surface.

Thanks!!!

And, Sweet Dreams